# Supplementary Information

DeSurv identifies a replicable tumor–stroma prognostic architecture in pancreatic cancer through survival-supervised matrix factorization

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## 1 Model details

Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix with $p$ genes and $n$ subjects. DeSurv approximates $X \approx WH$ with $W \in \mathbb{R}_{\geq 0}^{p\times k}$ and $H \in \mathbb{R}_{\geq 0}^{k\times n}$, and links the shared program matrix $W$ to survival via an elastic-net–penalized Cox model. Supervision is placed on the program matrix $W$, the basis that transfers to new cohorts, whereas the sample loadings $H$ serve only to reconstruct the training expression matrix; new samples are therefore scored by projection onto the learned programs, $Z = W^\top X$, rather than by re-estimating loadings.

Let $y \in \mathbb{R}_{>0}^{n}$ be the observed survival times, $\delta \in \{0,1\}^n$ the event indicators, and $Z = W^\top X \in \mathbb{R}^{k\times n}$ the sample-wise factor scores. We write $Z_i$ for the $i$th column of $Z$, and $n_{\text{event}} = \sum_{i=1}^{n} \delta_i$. For convenience we recall the DeSurv loss from Equation 1 in the main text and rewrite it as

$$
\begin{aligned}
\mathcal{L}(W,H,\beta) = &\\
&(1-\alpha)\left(\frac{1}{2\|X\|_F^2}\|X-WH\|_F^2 + \frac{\lambda_H}{2nk}\|H\|_F^2\right)\\
&-\alpha\left(\frac{2}{n_{event}}\ell(W,\beta) - \lambda\{\xi\|\beta\|_1 + \tfrac{1-\xi}{2}\|\beta\|_2^2\}\right)
\end{aligned}
$$

where $\|\cdot\|_F$ is the Frobenius norm and $\beta \in \mathbb{R}^k$ are the Cox coefficients. The Cox log-partial likelihood $\ell(W,\beta)$ is

$$
\ell(W,\beta) = \sum_{i=1}^{n} \delta_i \left[ Z_i^\top \beta - \log\left( \sum_{j: y_j \geq y_i} \exp(Z_j^\top \beta) \right) \right]. \tag{S1}
$$

The inner sum runs over the risk set $\{j : y_j \geq y_i\}$ of subjects still at risk at the observed time $y_i$; tied event times are handled by the Breslow approximation in the `glmnet` Coxnet routine used for the $\beta$ update (below). Hyperparameters satisfy $\alpha \in [0,1)$, $\lambda_H \geq 0$, $\lambda \geq 0$, and $\xi \in [0,1]$. The constants $1/(2\|X\|_F^2)$, $2/n_{\text{event}}$, and $1/(2nk)$ place the reconstruction, loadings-penalty, and survival terms on comparable numerical scales for stable optimization. The elastic-net mixing parameter $\xi$ is the argument `nu` in the DeSurv package.

## 2 Optimization Algorithm

### 2.1 Block coordinate descent scheme

Optimization proceeds by block coordinate descent (BCD; Algorithm S1), which at each outer iteration updates $H$, then $W$, then $\beta$ while holding the other blocks fixed.

### 2.2 Update for $H$

Conditional on $W$, the loss $\mathcal{L}(W,H,\beta)$ reduces to a convex quadratic function of $H$. We adopt the standard NMF multiplicative update with $\ell_2$ penalty:

$$
H \leftarrow \max\left( H \odot \frac{W^\top X}{W^\top WH + \frac{\|X\|_F^2}{nk}\,\lambda_H H + \varepsilon_H}, \varepsilon_H \right), \tag{S2}
$$

**Algorithm S1** DeSurv block coordinate descent

**Input:** $X \in \mathbb{R}_{\geq 0}^{p \times n}$, survival times $y \in \mathbb{R}_{>0}^{n}$, event indicators $\delta \in \{0,1\}^n$, tolerance $tol$, maximum iterations $maxit$, supervision parameter $\alpha$, rank $k$, penalties $\lambda_H$, $\lambda$, and $\xi$

**Output:** Fitted DeSurv parameters $(W, H, \beta)$

Initialize $W^{(0)}, H^{(0)}$ with positive entries (e.g., $W^{(0)}, H^{(0)} \sim \text{Uniform}(0, \max X)$)
Initialize $\beta^{(0)} = \mathbf{0}$ (the value used in all reported fits)
$loss = \mathcal{L}(W^{(0)}, H^{(0)}, \beta^{(0)})$
$eps = \infty$, $t = 0$
**while** $eps \geq tol$ **and** $t < maxit$ **do**
    **(H-update)**
        Update $H^{(t+1)}$ from $H^{(t)}$ using the multiplicative rule in Eq. S2,
        holding $W^{(t)}$ and $\beta^{(t)}$ fixed.
    **(W-update)**
        Update $W^{(t+1)}$ from $W^{(t)}$ using the hybrid multiplicative rule in Eq. S6,
        holding $H^{(t+1)}$ and $\beta^{(t)}$ fixed, and perform backtracking to ensure
        $\mathcal{L}$ does not increase.
    **($\beta$-update)**
        Update $\beta^{(t+1)}$ by solving the elastic-net Cox subproblem using cyclic coordinate descent to numerical tolerance
        (Eq. S7), holding $W^{(t+1)}$ and $H^{(t+1)}$ fixed.
    $lossNew = \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)})$
    $eps = |lossNew - loss|/|loss|$
    $loss = lossNew$
    $t = t + 1$
**end while**
**return** $\hat{W} = W^{(t)}, \hat{H} = H^{(t)}, \hat{\beta} = \beta^{(t)}$

where $\odot$ denotes elementwise multiplication, all divisions are elementwise, and $\varepsilon_H > 0$ is a small floor to prevent entries from becoming exactly zero. This update is equivalent to a majorization–minimization step and guarantees a nonincreasing reconstruction term conditional on $W$[1,2]. The $\varepsilon_H$-floor ensures that iterates remain in the interior of the nonnegative orthant, which keeps the multiplicative $H$-update numerically well-defined.

## 2.3 Update for $W$

For $W$, we construct a hybrid multiplicative update that balances the contributions of the NMF and Cox gradients. Let

$$\nabla_W \mathcal{L}_{\text{NMF}}(W, H) = \frac{1}{\|X\|_F^2}(WH - X)H^\top,$$

and let

$$\nabla_W \mathcal{L}_{\text{Cox}}(W, \beta) = \frac{2}{n_{event}} \nabla_W \ell(W, \beta),$$

where $\nabla_W \ell(W, \beta)$ denotes the Cox gradient with respect to $W$. A closed-form expression for $\nabla_W \ell$ is

$$\nabla_W \ell(W, \beta) = \sum_{i=1}^{n} \delta_i \left( x_i - \frac{\sum_{j: y_j \geq y_i} x_j \exp(x_j^\top W \beta)}{\sum_{j: y_j \geq y_i} \exp(x_j^\top W \beta)} \right) \beta^\top, \tag{S3}$$

where $x_i$ denotes the $i$th column of $X$.

We define a gradient-scaling heuristic

$$\zeta^{(t)} = \frac{\left\|(1-\alpha)\nabla_W \mathcal{L}_{\text{NMF}}(W^{(t)}, H^{(t+1)})\right\|_F + \varepsilon_\zeta}{\left\|\alpha \nabla_W \mathcal{L}_{Cox}(W^{(t)}, \beta^{(t)})\right\|_F + \varepsilon_\zeta}, \tag{S4}$$

with a numerical stabilizer $\varepsilon_\zeta = 10^{-12}$ in each norm, and clip $\zeta^{(t)}$ to avoid extreme values: $\zeta^{(t)} \leftarrow \min(\zeta^{(t)}, \zeta_{\max})$ with $\zeta_{\max} = 10^6$ in all experiments. The scaling removes differences in the magnitudes of the weighted reconstruction and survival gradients before the additional factor $\alpha$ below controls the relative Cox contribution. Because the Cox gradient entering $\zeta$ is already weighted by $\alpha$, the additional explicit factor of $\alpha$ in the $W$ update below makes the scaled Cox contribution approximately $\alpha$ times the weighted reconstruction-gradient norm when the cap is inactive, up to the numerical stabilizer; the two factors of $\alpha$ in that update are therefore not a duplication.

In the reported consensus fit, $\zeta$ decreased from 0.71 after the first iteration to 0.059 at convergence, and the scaled Cox contribution was then 1.6% of the reconstruction term in the numerator. The cap was reached only at initialization, where $\beta = 0$ makes the Cox gradient itself zero, so the cap contributed no update; across the 100 single-run endpoints $\zeta$ ranged from 0.003 to 0.031 and was never capped (the $\zeta$-trace script, Table S4, reconstructs the production optimization path from the cached seed fits and checks it against the cached final fit to a declared tolerance).

The multiplicative factor for $W$ is then

$$R^{(t)} = \frac{\frac{(1-\alpha)}{\|X\|_F^2} X H^{(t+1)\top} + \alpha\, \zeta^{(t)} \, \frac{2\alpha}{n_{\text{event}}} \, \nabla_W \ell(W^{(t)}, \beta^{(t)})}{\frac{(1-\alpha)}{\|X\|_F^2} W^{(t)} H^{(t+1)} H^{(t+1)\top} + \varepsilon_W}, \tag{S5}$$

$R^{(t)}$ is clamped elementwise to $[0.2, 5]$ before use, which bounds the change of any entry of $W$ in a single step, and the proposed update is

$$W^{(t+1)} = \max\Big(W^{(t)} \odot R^{(t)}, \varepsilon_W\Big), \tag{S6}$$

with a small floor $\varepsilon_W > 0$. When $\alpha = 0$, this reduces to the standard multiplicative update for NMF[1]; for $\alpha > 0$, the Cox gradient perturbs the update in a direction that decreases the supervised loss.

Because the combined multiplicative step does not, by itself, guarantee a nonincrease in the full loss $\mathcal{L}$, we embed it in a backtracking line search.

**Algorithm S2** $W$ update with backtracking

**Input:** $W$ at the previous iteration $t$, $W^{(t)}$; the multiplicative update term at the current iteration $R^{(t)}$, the max number of backtracking updates $max\_bt$, the initial step $s_0 \in (0, 1]$, the backtracking factor $\gamma \in (0, 1)$

**Output:** $W^{(t+1)}$

$s = s_0$
b = 1
$flag\_accept = FALSE$
**while** $b \leq max\_bt$ **do**
  $W^{(t+1)}_{cand} = \max\big(W^{(t)} \odot [(1-s) + sR^{(t)}],\ \varepsilon_W\big)$
  **(Column normalization)**
    Compute $D = \mathrm{diag}\big(\|W^{(t+1)}_{(:,1)cand}\|_2, \ldots, \|W^{(t+1)}_{(:,k)cand}\|_2\big)$
    and set $W^{(t+1)}_{cand} \leftarrow W^{(t+1)}_{cand} D^{-1}$, $H^{(t+1)}_{cand} \leftarrow DH^{(t+1)}$, and $\beta^{(t)}_{cand} \leftarrow D\beta^{(t)}$
  **if** $\mathcal{L}(W^{(t+1)}_{cand}, H^{(t+1)}_{cand}, \beta^{(t)}_{cand}) \leq \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)})$ **then**
    $W^{(t+1)} = W^{(t+1)}_{cand}$
    $H^{(t+1)} = H^{(t+1)}_{cand}$
    $\beta^{(t)} = \beta^{(t)}_{cand}$
    $flag\_accept = TRUE$
    break
  **end if**
  $s = s\gamma$
  $b = b + 1$
**end while**
**if** $flag\_accept = FALSE$ **then**
  $W^{(t+1)} = W^{(t)}$
**end if**

In all reported fits $s_0 = 0.5$, $\gamma = 0.5$ and $max\_bt = 10$ (the values passed by the package's R wrapper to the optimizer). The column normalization inside the line search is described in Normalization of $W$ below; it is applied to each candidate before the objective is compared, so the accepted update is the normalized one.

## 2.4 Update for $\beta$

Conditional on $(W, H)$, the loss in $\beta$ reduces to a convex elastic-net–penalized Cox problem; the common factor $\alpha > 0$ multiplies both the log-partial-likelihood and the penalty in the joint loss (main-text Equation 1) and is dropped here, as it does not affect the minimizer:

$$\min_{\beta \in \mathbb{R}^k} \left\{ -\frac{2}{n_{\text{event}}}\, \ell(W, \beta) + \lambda \left( \xi \|\beta\|_1 + \frac{1-\xi}{2} \|\beta\|_2^2 \right) \right\}.$$

We solve this subproblem by cyclic coordinate descent following[3]. Writing $\ell(\beta) = \ell(W, \beta)$ and $\tilde{\eta}_i = Z_i^\top \beta$, the update for coordinate $r$ has the closed form

$$\hat{\beta}_r = \frac{S\left( \dfrac{1}{n_{\text{event}}} \sum_{i=1}^n w(\tilde{\eta})_i z_{i,r} \left[ v(\tilde{\eta})_i - \sum_{j \neq r} z_{i,j} \beta_j \right], \lambda\xi \right)}{\frac{1}{n_{\text{event}}} \sum_{i=1}^n w(\tilde{\eta})_i z_{i,r}^2 + \lambda(1-\xi)}, \tag{S7}$$

where $S(a, \tau) = \mathrm{sign}(a) \max(|a| - \tau, 0)$ is the soft-thresholding operator, and $w(\tilde{\eta})$, $v(\tilde{\eta})$ are standard local quadratic approximations of the Cox log-partial likelihood[3]; the factor $1/n_{\text{event}}$ arises because the $2/n_{\text{event}}$ weight on the log-partial likelihood in the objective cancels the one-half of the quadratic approximation. We iterate coordinate updates until the subproblem converges to numerical tolerance, yielding a minimizer in $\beta$ for the current $W$.

### 2.5 Normalization of W

After each accepted $W$ update, we normalize the columns of $W$ as $W \leftarrow WD^{-1}$, and adjust $H$ and $\beta$ as

$$H \leftarrow DH, \qquad \beta \leftarrow D\beta,$$

where $D$ is the diagonal matrix of column $\ell_2$-norms of $W$. This preserves the reconstruction $WH$ and the linear predictor $W\beta$; because the accompanying rescaling of $H$ and $\beta$ can alter the penalization terms, the normalized proposal is evaluated under the full penalized objective within the backtracking procedure for the $W$ update rather than assumed loss-invariant. Normalization prevents degeneracy in the scale-nonidentifiable factorization and keeps columns of $W$ comparable in magnitude and interpretable as gene programs.

# 3 Supervision on $W$ versus $H$

Classical nonnegative matrix factorization (NMF) is non-identifiable: for any invertible matrix $R$ with $WR \geq 0$ and $R^{-1}H \geq 0$, the factorization $(W, H)$ can be transformed to $(WR, R^{-1}H)$ without changing the reconstruction error, yielding multiple valid solutions[4–6]. Although normalizing columns of $W$ or rows of $H$ resolves the trivial scaling ambiguity, nonnegative mixing transformations persist, so factorizations started from different initializations need not agree[7–9]. This applies to DeSurv as well: individual restarts of the reported model are dispersed, and the final fit is obtained by a consensus procedure over many restarts (Stability of the recovered gene programs).

In DeSurv, survival supervision enters through the projection $Z = W^\top X$ in the partial log-likelihood, so that the gradient of the DeSurv loss with respect to the programs is

$$\nabla \mathcal{L} = (1-\alpha)\nabla_W \mathcal{L}_{\text{NMF}}(W, H) - \alpha \nabla_W \mathcal{L}_{\text{Cox}}(W, \beta),$$

so the update for $W$ in Equation S6 combines the reconstruction and supervision terms, and the gene-program definitions are shaped directly by their association with survival.

By contrast, if supervision were applied to the sample loadings $H$, the model could reduce the Cox loss by redistributing patient-specific coefficients while leaving the gene-level programs $W$ nearly unchanged, a behavior documented for supervised topic models, where supervision applied only to the coefficient matrix modifies $H$ but not the basis $W$[10].

Supervising $W$ also provides a practical advantage for **portability**: because $W$ defines gene-level programs, new samples can be scored directly by the projection $Z_{\text{new}} = W^\top X_{\text{new}}$. Supervising $H$ would not yield such a mapping: obtaining sample loadings for external datasets would require an additional sample-specific estimation step, a nonnegative least-squares (NNLS) problem for each new sample.

Supervising through $W$ is therefore intended to shape the gene programs toward prognostic directions and to yield signatures that can be scored unchanged in new cohorts; whether they do so is an empirical question addressed in Part II.

Supplementary Table S1 places DeSurv beside the unsupervised factorization it extends, the two published survival-supervised factorizations, and the two conventional supervised predictors used as comparators, by where survival enters each method, what object is learned, and what must happen when a new sample arrives. It records architecture only; it does not rank the methods.

# 4 Cross-validation procedure

Algorithm S3 describes the cross validation procedure with $F$ folds and $R$ initializations. We define the C-index using comparable pairs $\mathcal{P} = \{(i, j) : y_i < y_j,\ \delta_i = 1\}$ and linear predictors $\hat{\eta}_i$:

$$\hat{c} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left[ \mathbb{1}(\hat{\eta}_i > \hat{\eta}_j) + \tfrac{1}{2}\mathbb{1}(\hat{\eta}_i = \hat{\eta}_j) \right]. \tag{S8}$$

**Algorithm S3** Cross-validation for DeSurv pipeline

**Input:**

Number of folds $F$
Number of initializations $R$
Observed data $(X,\, y,\, \delta)$
Hyperparameters $k$, $\alpha$, $\lambda$, $\xi$, $\lambda_H$, and number of top genes per factor $n_{top}$

**Output:** The cross-validated C-index

```
Divide subjects into F folds.
for f = 1 to F do
    Split data into training and validation (X_(-f), y_(-f), δ_(-f)), and (X_(f), y_(f), δ_(f)) sets
    for r = 1 to R do
        set.seed(r)
        Apply Algorithm S1 with inputs (X_(-f), y_(-f), δ_(-f)), and (k, α, λ, ξ, λ_H)
        Obtain Ŵ and β̂ as output from Algorithm S1
        Restrict Ŵ to the top n_top genes per factor (by factor-specificity score) and compute the estimated
    linear predictor on this support: η̂ = X_(f)^T Ŵ β̂
        Compute C-index, denoted ĉ_(f)r, according to Equation S8
    end for
    end loop over r
    Compute average C-index across initializations: ĉ_(f) = (1/R) Σ_{r=1}^{R} ĉ_(f)r
end for
end loop over f
Compute final cross-validated C-index: ĉ = (1/F) Σ_{f=1}^{F} ĉ_(f)
return Final estimate ĉ
```

# 5 Bayesian Optimization

We treat hyperparameter tuning as a black-box optimization problem over a bounded domain. Let $\psi$ collect the tuned parameters (at minimum $\psi = (k, \alpha, \lambda, \xi)$, and optionally discrete choices such as the signature size $n_{\text{top}}$). For a given $\psi$ we run the cross-validation procedure in Algorithm S3, averaging the C-index across folds and random initializations to form the objective

$$\hat{c}(\psi) = \frac{1}{F} \sum_{f=1}^{F} \hat{c}_{(f)}(\psi).$$

Because $\hat{c}(\psi)$ is expensive to evaluate and non-differentiable with respect to $\psi$, we use Bayesian optimization (BO) to adaptively select candidate configurations.

We place a Gaussian-process (GP) prior on the response surface $f(\psi) \sim \mathcal{GP}(0, k(\psi, \psi'))$ with a Matérn kernel and automatic relevance determination length-scales. Continuous parameters are rescaled to $[0, 1]$ and penalties are optimized on a log scale; integer-valued coordinates (e.g., $k$ or $n_{\text{top}}$) are proposed in the continuous space and rounded inside the evaluator. Following an initial batch of evaluations, the GP is refit each time a new observation is added and used to guide acquisition.

At iteration $t$, let $\mu_t(\psi)$ and $\sigma_t(\psi)$ denote the GP posterior mean and standard deviation, and let $f_\star$ be the best observed score. We select new candidates by maximizing expected improvement (EI),

$$\text{EI}(\psi) = (\mu_t(\psi) - f_\star - \kappa)\, \Phi(Z(\psi)) + \sigma_t(\psi)\phi(Z(\psi)), \quad Z(\psi) = \frac{\mu_t(\psi) - f_\star - \kappa}{\sigma_t(\psi)},$$

where $\Phi$ and $\phi$ are the standard normal cdf/pdf and $\kappa \geq 0$ controls the exploration margin. EI is evaluated over a candidate pool sampled from the bounded search region, and the maximizer is evaluated next. The loop continues until the iteration budget is exhausted or the improvement plateaus.

For the PDAC application, the objective was the mean 5-fold cross-validated Harrell C-index (Algorithm S3), with five-fold cross-validation constructed separately within cohort and stratified by event status under a fixed seed (123); each configuration was evaluated from 30 random initializations and its C-index averaged across them. The bounded search domain was $k \in [2, 12]$, $\alpha \in [0, 1]$ (the proposal bounds; the model domain is $\alpha \in [0, 1)$, package validation clips proposals at $\alpha \geq 1$, and the largest evaluated value was 0.986), $\lambda \in [10^{-3}, 10^3]$ (optimized on a $\log_{10}$ scale), $\xi \in [0, 1]$, and $n_{\text{top}} \in [50, 300]$, with the loadings ridge penalty fixed at $\lambda_H = 0$. Bayesian optimization used an initial design of 50 evaluations followed by 100 GP-guided iterations (150 total evaluations), maximizing expected improvement over a candidate pool of 4,000 points sampled from the bounded domain at each step with an exploration margin $\kappa = 0.01$. The unsupervised comparator fixed $\alpha = 0$ and tuned the remaining coordinates identically.

For model selection, we take the highest observed $\hat{c}(\psi)$ as the optimal configuration. When fold-level standard errors are available, we apply a one-standard-error rule for $k$: for each rank we retain the configuration with the highest observed mean cross-validated C-index, set the threshold one fold-level standard error below the overall best configuration's mean, and select the smallest rank whose retained configuration meets it.

Two details of this rule matter for reading the reported numbers. First, the comparison is between the best configuration observed at each rank: the highest mean cross-validated C-index anywhere in the search was 0.655 at $k = 7$ (fold-level standard error 0.011), and the best configuration observed at $k = 3$ reached 0.646, above the threshold of 0.644. Second, the final model is a hybrid of the two: the rule fixed $k = 3$, while the remaining hyperparameters ($\alpha = 0.334$, $\lambda = 0.349$, $\xi = 0.056$, $n_{\text{top}} = 270$) were retained from the overall best configuration at $k = 7$. The final configuration was therefore not itself a point evaluated in the search, and neither number above is its cross-validated performance. The nearest evaluated point is in the fixed-hyperparameter sensitivity grid (Cross-validated C-index across factorization rank), which holds the same $\lambda$, $\xi$ and $n_{\text{top}}$: at $k = 3$, $\alpha = 0.35$, the five-fold cross-validated C-index was 0.656 (standard error 0.010). The final DeSurv model is refit at the selected parameters using the consensus-based initialization (Consensus initialization for the final model), and the BO-chosen $n_{\text{top}}$ is used to truncate each program for downstream analysis.

The parameter $n_{\text{top}}$ determines the factor-specific gene support used for projection and, in the final consensus procedure, the gene sets from each restart that enter the consensus initialization. After factorization, each sample's score for factor $j$ is computed as $\widetilde{W}_j^\top x$, where $\widetilde{W}_j$ retains only the $n_{\text{top}}$ genes with the largest factor-specificity scores $s_{ij}$ (Supplementary Information, Section 8); genes outside this set still shape the factorization but do not enter the linear predictor. BO selects $n_{\text{top}}$ jointly with $k$, $\alpha$, $\lambda$, and $\xi$ by optimizing cross-validated C-index. When $\alpha = 0$ (no survival supervision), survival does not enter the factorization objective; it is used for the post hoc Cox fit and for tuning $n_{\text{top}}$, which sets the projection support and, through the consensus initialization, can still influence the final unsupervised fit indirectly.

Supplementary Fig. S1 illustrates the tuned response surface for the two coordinates of primary interest in the PDAC training cohort: the cross-validated C-index over factorization rank $k$ and supervision strength $\alpha$. Performance is high across a broad region at intermediate supervision and moderate rank, with a second high region at $k \geq 9$ and $\alpha \approx 0.2$–$0.3$, and degrades toward the unsupervised corner ($\alpha = 0$), consistent with a benefit of survival supervision; the second region is consistent with the flat rank dependence of the supervised curve in Section 23. This is an empirical model-selection sensitivity surface, not a simulation result.

## 5.1 Consensus initialization for the final model

The final DeSurv model at the selected hyperparameters was initialized from a consensus of multiple independent runs to reduce sensitivity to random starts. We first fit $R = 100$ DeSurv models to the training data from independent random nonnegative initializations at the selected $(k, \alpha, \lambda, \xi, n_{\text{top}})$ (seeds 1–100). For each run, the top $n_{\text{top}}$ genes of every learned program were extracted by factor-specificity score, and a gene-by-gene consensus matrix was accumulated across runs, recording how often each pair of genes appeared together among the top genes of the same program.

Genes appearing in the top sets of fewer than $\lceil 0.3R \rceil$ of the $R$ runs were treated as inactive and excluded from clustering. Because the consensus is constructed from gene-by-gene co-occurrence across restarts, it is invariant to factor-label permutations and does not require explicit alignment of factor labels across runs. The consensus matrix over the remaining genes was partitioned into $k$ groups by agglomerative hierarchical clustering (average linkage), and each cluster defined one column of the initial gene-program matrix $W_0$; inactive genes and otherwise-empty entries were set to a small positive floor ($\sqrt{\epsilon_{\text{mach}}}$) to preserve strict positivity. The initial loadings $H_0$ were obtained by nonnegative least squares given $W_0$ and the training expression matrix, and the initial Cox coefficients $\beta_0$ were set to zero. This consensus seed used only training-cohort data and was applied solely to initialize the single final refit after hyperparameter selection; it used no validation-cohort data and therefore introduced no information leakage.

# 6 PDAC Datasets

We analyzed PDAC cohorts from TCGA-PAAD and CPTAC-PDAC as training data[11,12]. External validation used independent cohorts from Dijk, Moffitt, PACA-AU (array and RNA-seq), and Puleo[13–16]. Training models were fit on the combined TCGA and CPTAC expression matrices after harmonizing gene identifiers; validation datasets were processed with the same gene set and transformations before projection, scoring and downstream validation analyses.

Dataset-specific inclusion and preprocessing steps were as follows. For TCGA-PAAD, we excluded samples lacking tumor grade and mapped features to gene symbols. For CPTAC, we retained samples annotated as PDAC by histology. For Dijk, all 90 samples passed quality-control filters and were retained. For Moffitt, only primary tumor specimens were kept. For PACA-AU array and RNA-seq, we retained primary PDAC tumors and collapsed duplicated gene symbols by median. Puleo required no additional dataset-specific filtering beyond survival QC.

Across cohorts, samples without valid survival outcomes (nonpositive time, missing time, or missing event indicator) were removed. Expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort (TCGA-PAAD and CPTAC) separately, genes were ranked in descending order of mean expression and, independently, of variance; the 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection of these two gene sets yielded 1,970 genes used for model training. Validation datasets were restricted to this same 1,970-gene set.

A within-sample rank transformation was then applied to every dataset to mitigate platform and scale effects. Specifically, for each sample the expression values across the 1,970 retained genes were replaced by their ascending ranks (ties resolved by average ranks), so that every sample carries the same nonnegative rank distribution over the shared gene set; ranks are inherently nonnegative and therefore preserve NMF compatibility. Validation cohorts were rank-transformed by exactly the same per-sample procedure before projection, with no use of validation outcomes.

Table S2 summarizes the cohorts, platforms, sample sizes after quality control, and patient-level metadata used in this study. Overall survival time and event indicators were available for all cohorts. PurIST[17] and DeCAF[18] molecular classifications, applied to bulk expression data as described in those references, were used as covariates in the covariate-adjusted external validation analysis. Gene expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study[18]; training data were obtained directly from GDC.

# 7 Survival Analysis

Survival analyses used overall survival time with event indicators as provided by each cohort. For fitted DeSurv models, sample-level program scores are the raw projections $Z = \widetilde{W}^\top X$ onto the frozen top-$n_{\text{top}}$ support, and the linear predictor is $\hat{\eta} = Z\hat{\beta}$ with the trained coefficients; no re-estimation and no standardization enters the score itself. Throughout, projection denotes multiplication by this fixed gene-weight matrix; the truncated columns are not renormalized, so the term does not imply an orthogonal geometric projection. The

Cox coefficients $\hat{\beta}$ were estimated during model fitting from the full-support scores $W^\top X$ over all analyzed genes; for cross-validation and external scoring they were carried forward unchanged and applied to scores from the truncated basis $\widetilde{W}$, and were not refit after truncation (Algorithm S3). The elastic-net penalty set the trained coefficient for D2 to exactly zero, so $\hat{\eta}$ is a weighted sum of the D1 and D3 scores. Prognostic performance was summarized using the concordance index as defined above (Cross-validation procedure). When multiple datasets were pooled, Cox models were stratified by dataset to allow cohort-specific baseline hazards.

Every reported linear predictor is this same linear functional of expression up to a positive constant; the manuscript uses three scalings of it, and the map from scaling to reported quantity is as follows. (i) During cross-validated tuning the collapsed weight vector $\widetilde{W}\hat{\beta}$ is scaled to unit $\ell_2$ norm (main text, Methods); this changes $\hat{\eta}$ by a positive constant only and leaves every concordance index unchanged. (ii) The headline pooled hazard ratio, the covariate-adjusted hazard ratios (main-text Table 1) and the leave-one-dataset-out estimates divide $\hat{\eta}$ by its standard deviation over the pooled, de-duplicated validation set, so they are per pooled standard deviation. (iii) The per-factor forest estimates (main-text Fig. 3a), the cohort-specific hazard ratios of Table S9 and the between-cohort heterogeneity test standardize each score to zero mean and unit standard deviation within its own cohort, so they are per within-cohort standard deviation. Hazard ratios under (ii) and (iii) are therefore not interchangeable, whereas C-indices are identical under all three.

### 7.1 Independence of discovery and validation

DeSurv is trained with survival supervision only; it never uses PurIST, DeCAF, or any molecular subtype label during factorization or hyperparameter selection. The PurIST and DeCAF subtype calls were fixed before the DeSurv analysis (loaded as precomputed per-cohort annotations) and enter only as covariates in the adjusted Cox models, not as inputs to DeSurv. All rank and hyperparameter selection (including the one-standard-error rule) was performed by cross-validation entirely within the TCGA and CPTAC training data; no external validation cohort was used for training or tuning. Validation-cohort expression enters only at scoring, a fixed-basis projection ($Z = \widetilde{W}^\top X_{\text{new}}$) that uses no validation outcome, and validation-cohort survival outcomes were used only for the final, out-of-sample performance evaluation.

### 7.2 Cohort handling in the training objective

The training Cox objective was fit to the pooled TCGA-PAAD and CPTAC samples without a cohort term; cross-validation folds were constructed within cohort with stratification by event status. Because survival shapes $W$, a cohort effect associated with both expression and survival could in principle be rewarded by the survival gradient. As a descriptive check for cohort-driven association, computed from the cached training data and final fit and not a refit, the fitted linear predictor remained strongly associated with survival within TCGA-PAAD and CPTAC separately (in-sample HR per SD 3.50 and 3.72; Table S7) and in a model stratified by cohort (HR per SD 3.35, against 3.12 unstratified), while the between-cohort shift in the score was opposite to the between-cohort hazard difference: the predictor averaged 0.73 SD higher in TCGA-PAAD, yet CPTAC had the higher hazard given the predictor (HR 1.90). These in-sample quantities argue against the specific mechanism in which the survival gradient learns cohort membership, but they do not rule out cohort influence on the factorization, which the external validation addresses directly.

## 8 Gene Program Correlation Analysis

To assess the biological identity of learned factors, we evaluated the correspondence between factor-specific gene rankings and established PDAC gene programs. Each column $j$ of the gene program matrix $W$ was first scaled by its column maximum, $W^*_{ij} = W_{ij} / \max_i W_{ij}$, placing all factors on a common $[0, 1]$ scale, and the specificity score for gene $i$ in factor $j$ was defined as

$$s_{ij} = W^*_{ij} - \max_{j' \neq j} W^*_{ij'}.$$

Genes were ranked in descending order of $s_{ij}$. This ranking defines both the top-$n_{\text{top}}$ genes used to score samples (Supplementary Data: factor-specific gene lists) and, for the top 50 genes per factor, the loadings used to quantify factor–program correspondence.

For each factor $j$ and each reference gene program $G_k$, we computed the Spearman rank correlation between the continuous factor loadings $W_{\cdot j}$ (restricted to the union of the per-factor top-50 gene sets intersected with the reference programs) and a binary indicator vector $\mathbf{1}[\text{gene} \in G_k]$ (main-text Fig. 2a,b; Methods). $P$ values were derived from the asymptotic Spearman test and adjusted for multiple comparisons using the Benjamini–Hochberg procedure within each factor column independently; as in the main text, these adjusted $P$ values are displayed as descriptive marks, because the gene universe is itself selected by loading. Reference programs were included in the visualization only if at least one factor yielded a Spearman correlation exceeding 0.2.

The reference universe comprises 45 published PDAC gene programs (Table S3): the PurIST basal-like and classical tumor signatures[17]; the DECODER compartment signatures[19]; the Moffitt tumor, stroma and immune programs[14]; the Collisson subtypes[20]; the Bailey subtypes, including ADEX (aberrantly differentiated endocrine exocrine)[15]; the Chan-Seng-Yue subtypes[21]; the Maurer ECM and immune programs[22]; the five Puleo whole-tumor components together with Puleo's two tumor-compartment signatures[16]; the Elyada iCAF and myCAF signatures[23]; the SCISSORS fibroblast and tumor signatures[24]; and the DeCAF proCAF and restCAF programs[18]. Nine source entries were excluded before analysis: six that duplicate a retained list gene for gene (a second copy of the Moffitt activated and normal stroma lists, the SCISSORS CAF-versus-perivascular variants of its apCAF, iCAF and myCAF lists, and a second copy of its perivascular list), the two nine-gene PurISS iCAF and myCAF lists, and Bailey's non-unique gene set. The SCISSORS iCAF and myCAF signatures are displayed under the DeCAF nomenclature, restCAF and proCAF (Stromal resolution: which reference programs are recovered from bulk). Puleo's whole-tumor pure basal-like component and its tumor-compartment basal-like signature are distinct programs and are labeled as such in every figure. Many programs place only a minority of their genes inside the 1,970-gene analysis set; the correlation for a program is computed over the genes it shares with that set, and a program with no such genes cannot be represented.

# 9 Reconstruction and Survival Contribution per Factor

NMF does not provide an orthogonal variance decomposition; it seeks a low-rank reconstruction of the expression matrix $X \approx WH$. We therefore measure each factor's contribution to reconstruction rather than to variance explained. For a subset of factors $S \subseteq \{1, \dots, k\}$, define the reconstruction value

$$v(S) = \|X\|_F^2 - \Big\| X - \sum_{j \in S} W_{\cdot j} H_{j \cdot} \Big\|_F^2,$$

the reduction in residual error relative to the null (zero-factor) model, where $W_{\cdot j} H_{j \cdot}$ is the rank-one outer-product contribution of factor $j$ (the outer product of column $j$ of $W$ with row $j$ of $H$). The reconstruction Shapley value[25] of factor $j$,

$$\phi_j = \sum_{S \subseteq \{1,\dots,k\} \setminus \{j\}} \frac{|S|!\,(k - |S| - 1)!}{k!} \big( v(S \cup \{j\}) - v(S) \big),$$

is its average marginal reconstruction gain over all orderings of the factors.

By the Shapley efficiency axiom, $\sum_j \phi_j = v(\{1, \dots, k\})$, the total reconstruction gain of the full model, so the normalized shares $\phi_j / \sum_l \phi_l$ form an additive decomposition of reconstruction that sums to one; individual shares may be signed in principle but were all positive in the reported PDAC models. This is a genuine additive decomposition, in contrast to per-factor variance fractions, which for non-orthogonal factorizations such as NMF do not partition (cross-terms between rank-one contributions are non-zero). The Shapley shares are computed exactly by enumerating the $2^k$ subsets (trivial at the small ranks considered here).

The survival contribution of factor $j$ was quantified as the Type III partial likelihood increment: the reduction in Cox partial log-likelihood when factor $j$ is omitted from the full $k$-factor model,

$$\Delta\ell_j = \ell_{\text{full}}(\hat{\beta}_{\text{full}}) - \ell_{-j}(\hat{\beta}_{-j}),$$

where $\ell_{\text{full}}(\hat{\beta}_{\text{full}})$ is the maximized unpenalized Cox partial log-likelihood of the full $k$-factor model and $\ell_{-j}(\hat{\beta}_{-j})$ is that of the reduced model refit on the remaining $k-1$ factors after omitting factor $j$, each evaluated at its own maximum partial likelihood estimate. Larger $\Delta\ell_j$ indicates a greater conditional contribution of factor $j$ to Cox model fit given the remaining factors. Unlike the reconstruction Shapley shares, these Type III increments are conditional (each is evaluated against the full complement of the remaining $k-1$ factors) and are nonadditive: they do not partition the total model partial likelihood and are not comparable across factors as variance-style shares. We therefore report them as per-factor conditional increments rather than as an additive survival decomposition.

These two quantities are plotted against each other in main text, Fig. 2c. In PDAC at $k = 3$, the reconstruction shares are 41.8%/18.6%/39.7% for NMF (N1/N2/N3) and 18.8%/25.2%/56.0% for DeSurv (D1/D2/D3), while the survival contribution $\Delta\ell$ concentrates in DeSurv D1 ($\Delta\ell = 66.4$) with all NMF factors small ($\Delta\ell \leq 1.2$).

# 10 Factor Correspondence Analysis

To assess which biological programs are preserved, altered, or suppressed under survival supervision, we quantified the pairwise correspondence between NMF and DeSurv factor loadings. For each pair of factors $(j_{\text{NMF}}, j_{\text{DeSurv}})$, we computed the Spearman rank correlation between the corresponding columns of the NMF gene program matrix $W_{\text{NMF}}$ and the DeSurv gene program matrix $W_{\text{DeSurv}}$ over all genes. We use the full loading profile rather than a top-gene subset, which would impose a threshold and overstate apparent preservation. Spearman correlation was used to capture monotone correspondence in gene loading ranks while remaining robust to scale differences between the two methods. Results are displayed as a heatmap with NMF factors as rows and DeSurv factors as columns (main text, Fig. 2d).

# 11 Analysis scripts and cached objects

Numerical results from the analyses are read at render time from cached analysis objects rather than transcribed manually. Table S4 indexes the scripts referred to in this document by the analysis they perform, so that the prose above and below can name analyses rather than files.

# Part II

# Supplementary Results

## 12 Cross-method reconstructability and reconstruction fidelity

The factor-correspondence heatmap of Section 10 (main-text Fig. 2d) summarizes pairwise similarity; to make a direction-aware statement of how much of each factor is *linearly reconstructable* from the other method's basis, we regressed each factor's gene-loading vector onto the full set of the other method's loading vectors and report the ordinary least-squares $R^2$ (Table S5). A high $R^2$ means the factor is well reconstructed as a linear combination of the other method's factors; a low $R^2$ means it is not linearly reconstructable from that basis. This is a statement about linear reconstructability of gene-loading vectors, not about which method discovered a program first.

DeSurv D1 is reconstructed at $R^2 = 0.09$ from the NMF basis: its gene-loading vector is not well reconstructed as a linear combination of the matched-rank NMF factors. The biologically dominant tumor (N1) and stromal (N3) programs, by contrast, are largely reconstructable from the DeSurv basis ($R^2 = 0.90$ and 0.88), so most of the unsupervised structure is retained; the principal reorganization is that the DeSurv basis resolves tumor signal into basal- and classical-aligned programs while reconstructing the exocrine factor (N2) less well ($R^2 = 0.38$).

At the whole-matrix level, this reorganization carries a real but bounded reconstruction cost (Table S6). On the conventional centered $R^2$, DeSurv reconstructs 0.75 of the analysis matrix against 0.81 for unsupervised NMF, a difference of 6.0 percentage points and a relative loss of about 7%; this centered figure is the one reported in the main text. On the uncentered scale the gap is only 1.5 percentage points, but that scale flatters any nonnegative low-rank fit to a rank matrix.

## 13 The D1 axis is not specific to DeSurv

If the D1 tumor–stroma axis were an artifact of DeSurv's particular optimizer, independent survival-supervised methods should not recover it. We therefore fit three other supervised dimension-reduction methods on the same training cohort ($n = 273$, 139 events), namely supervised principal components (Cox-screened genes followed by PCA), one-component Cox partial least squares against martingale residuals, and a sparse Cox model (lasso-penalized; 32 genes selected in this single fit), and compared their leading survival-aligned direction to the DeSurv D1 score. All three recovered D1 closely (per-patient $|r| = 0.81$ for supervised PCA, 0.83 for Cox-PLS, and 0.85 for sparse Cox; mean 0.83; Fig. S2), and each aligned with D1 rather than with the stromal D2 or basal-like D3 programs (absolute correlations with D2 and D3 at most 0.19).

By contrast, the leading *unsupervised* direction (the first principal component) was only weakly correlated with D1 ($|r| = 0.10$) and instead tracked the high-variance D2/D3 stroma–basal axis. This D1 axis is thus shared across survival-supervised methods, not an idiosyncrasy of the DeSurv algorithm. Two additional findings are consistent with this conclusion: D1 is unreconstructable from the unsupervised NMF basis (Table S5), and Lee NMF at matched rank does not recover the restCAF and iCAF marker genes that appear among DeSurv's defining genes (Stromal resolution: which reference programs are recovered from bulk).

## 14 Robustness to tuned supervised comparators and compartment attribution

The preceding section establishes D1 as a survival-associated expression direction shared across supervised methods; here we ask what DeSurv adds beyond recovering that direction, using canonical, independently tuned comparators in place of the simple single-component versions above: supervised principal components (`superpc`; Bair–Tibshirani, with cross-validated screening threshold and component number) and lasso-penalized Cox regression (`glmnet`, mixing parameter 1, a single $\lambda_{\min}$ model). Cox partial least squares was not included in this comparison because a validated multi-component implementation was not available. All methods used the same 1,970-gene universe selected from the training data by expression and variance, and each performed its own feature selection; DeSurv was scored with its Bayesian-optimization-selected top-270-per-factor programs; all scoring used the frozen, rank-transformed transfer with no validation refitting.

External discrimination was similar: the pooled frozen-transfer C-index was 0.61 for the DeSurv linear predictor, 0.60 for supervised PCA, and 0.60 for lasso Cox. Consistent with the score-level recovery above, both tuned methods aligned with the D1 direction (training $|r| = 0.71$ for lasso Cox and 0.46 for supervised PCA) and with the DeSurv linear predictor ($|r| = 0.86$ for lasso Cox and 0.65 for supervised PCA), but were only weakly correlated with the activated-stroma/proCAF program D2 ($|r| = 0.06$ for lasso Cox and 0.14 for supervised PCA): a single supervised risk axis captures the dominant tumor-axis signal but not the separate stromal program.

The compartment attribution made this explicit. The genes emphasized by a supervised risk score need not coincide with D1's positively weighted defining genes, because the high-risk end of D1 is characterized in part by loss of classical-tumor and restCAF-like expression. Empirically they were basal-like tumor markers: across 12 cross-validation seeds the genes selected by both methods were enriched (hypergeometric $P < 0.01$) for the basal-like tumor program (in 12/12 seeds for lasso Cox and 12/12 for supervised PCA) and in no seed for the activated-stroma/proCAF, restraining-stroma, immune, or exocrine compartments. The supervised risk scores did nonetheless retain some stromal signal: each correlated modestly with an external activated-stroma/proCAF signature ($|r| = 0.20$ for lasso Cox and 0.23 for supervised PCA), but well below DeSurv's D2 program ($|r| = 0.55$), which tracks that signature directly.

The distinction is therefore representational: DeSurv preserves stromal context as an explicit program even when that program does not contribute independently to the fitted linear predictor.

# 15 Proper-scoring sensitivity of the external validation

The main-text external validation reports discrimination with Harrell's C-index under a Cox proportional-hazards model. Harrell's C ranks patients by a single time-invariant score, so when hazards are non-proportional, as the diagnostic below shows for this predictor, its value depends on the censoring distribution and need not reflect discrimination at any particular horizon[26,27], and it is not a proper scoring rule for predicted risks[28]. We therefore additionally evaluated the DeSurv linear predictor, its direction frozen from training, with a proper score, the integrated Brier score (IBS), in each validation cohort, recalibrating only its scalar slope and baseline hazard within each cohort and integrating over that cohort's follow-up to the 80th percentile of its event times.

Across the five validation platform datasets (PACA-AU array restricted to its array-only patients) the DeSurv predictor achieved an event-weighted integrated Brier score of 0.159 versus 0.168 for a Kaplan–Meier reference, an index of prediction accuracy ($\mathrm{IPA} = 1 - \mathrm{IBS}/\mathrm{IBS}_{\mathrm{null}}$) of 5.4% (per-cohort range 2.9–9.4%). Because each cohort's IBS is integrated over its own event-time horizon, this pooled value is a descriptive event-weighted summary rather than a single common-horizon proper score, and the per-cohort IPA range is the more directly interpretable quantity.

Two caveats limit the interpretation of this analysis. First, the recalibration step estimates both the scalar slope and the baseline hazard from each cohort's own outcomes, so unlike the C-index and hazard-ratio analyses, which use a fully frozen predictor, this sensitivity evaluates the transported score direction under local recalibration rather than a frozen absolute-risk model. Second, the scores are computed on the same data used to fit that recalibration, with no cross-validation or resampling, so the reported IBS and IPA are apparent in-sample values and overstate out-of-sample accuracy. Both caveats weaken the evidentiary value of the apparent improvement, which is already small. This modest proper-score improvement over the null is consistent with the moderate discrimination (C = 0.61) and with DeSurv's contribution being a separately resolved tumor- and stroma-associated decomposition rather than higher predictive accuracy.

# 16 Proportional-hazards diagnostic

Because the pooled hazard ratio assumes proportional hazards, we tested the scaled Schoenfeld residuals of the validation model, a Cox regression of overall survival on the standardized DeSurv linear predictor with a separate baseline hazard for each dataset (stratification), using the Grambsch–Therneau test. The proportional-hazards assumption was not supported for the standardized DeSurv linear predictor ($\chi^2 = 14.7$, df = 1, $P = 1.3 \times 10^{-4}$), with the association attenuating over follow-up: the estimated time-varying coefficient declines with follow-up (slope $-0.92$ on Kaplan–Meier-transformed time) from a positive value. The reported pooled hazard ratio is therefore a time-averaged summary of that association, not a constant multiplicative effect. The horizon-integrated proper-scoring (IBS/IPA) sensitivity above complements it by integrating prediction accuracy over each cohort's event-time horizon rather than through a single hazard-ratio summary.

# 17 Between-cohort heterogeneity and C-index confidence intervals

The five validation platform datasets span four independent patient cohorts, because the PACA-AU array and RNA-seq series profile a partially overlapping set of the same ICGC patients. All analyses in this section are patient-level and de-duplicated by the rule used throughout the manuscript: a patient profiled on both PACA-AU platforms is represented once, by the RNA-seq record, leaving 69 PACA-AU patients (17 array-only and 52 RNA-seq) and 570 unique patients with 388 deaths in total.

The DeSurv linear predictor was standardized *within* cohort (centered and scaled to unit standard deviation separately in each cohort), the same convention used for the per-factor forest estimates in main-text Fig. 3a and not the pooled-standard-deviation convention used for the headline pooled hazard ratio (Survival Analysis). Within-cohort standardization removes the scale degree of freedom (every cohort's score then has mean zero and unit standard deviation by construction), so a cohort-by-score interaction tests only whether the slope per within-cohort standard deviation differs between cohorts, and not whether the score is more dispersed in one cohort than another. Both models are Cox regressions of overall survival with a separate baseline hazard for each cohort (stratification): the common-slope model has the standardized predictor as its single covariate, and the cohort-specific-slope model adds the predictor-by-cohort interaction. Comparing them by likelihood-ratio test gave $\chi^2 = 1.47$ on 3 degrees of freedom ($P = 0.69$); standardizing within platform rather than within cohort, which removes the small residual location difference between the two PACA-AU platforms, gives $\chi^2 = 1.59$ on 3 degrees of freedom ($P = 0.66$).

Cochran's $Q$ over the four cohort-specific log hazard ratios and their standard errors was 1.47 on 3 degrees of freedom ($P = 0.69$), giving $I^2 = 0\%$ under $I^2 = \max\{0, (Q - \mathrm{df})/Q\}$. $Q$ and $I^2$ are descriptive summaries of dispersion among the four estimates and are reported as complements to the interaction test, not as a substitute for it. The likelihood-ratio test is the inferential statement, and with four cohorts both are low-powered against modest heterogeneity.

Confidence intervals for the C-index were obtained by a stratified nonparametric bootstrap on the same de-duplicated table, resampling patients with replacement *within* cohort so that no replicate mixes patients across cohorts or reinstates a de-duplicated PACA-AU record ($B = 1000$ replicates, seed 1, percentile intervals). Concordance was evaluated as Harrell's C with higher scores treated as higher risk, the convention of the external-validation C-indices, and the heterogeneity script (Table S4) verifies that this reproduces the per-dataset C-indices of Supplementary Table S8 exactly. The combined value is the event-weighted mean of the cohort C-indices, the same convention as the proper-scoring and tuned-comparator analyses, and therefore never forms a concordance pair across cohorts. The combined C-index was 0.613 (95% CI 0.579–0.646); the per-cohort hazard ratios and C-indices are given in Table S9.

## 18 Variance partition of the D1 score

The main text reports the fraction of the D1 score's variance explained by the binary PurIST and DeCAF classifications. D1 scores are first standardized within cohort, and an ordinary least-squares model is then fitted with platform dataset (five levels) as a fixed effect together with PurIST, DeCAF and their interaction. The quantity reported is the ordinary $R^2$ of that model, not the adjusted $R^2$; on $n = 570$ with eight estimated parameters the adjusted value is about one percentage point lower. Because the scores are standardized within platform dataset before fitting, the dataset main effect absorbs essentially no variance, so the within-dataset normalization of the reported statistic is a formality rather than a correction. The interaction term contributes negligibly, so the additive and interaction models give the same figure to the reported precision. Prognostic increments over the classifiers are assessed separately, by a partial likelihood-ratio test and the change in concordance from adding D1 to a cohort-stratified Cox model containing both classifiers.

D1 could also be a smoothed coding of the four joint PurIST-by-DeCAF subtype cells, so that its classical and restCAF-like arms would simply re-express one cell of a two-by-two. The main text reports the additive adjustment (D1 added to a cohort-stratified Cox model containing PurIST and DeCAF as main effects). As a sensitivity analysis we repeated that test with the PurIST-by-DeCAF interaction also in the base model; the statistics below are from this interaction model and differ slightly from the main-text additive ones for that reason. The interaction term itself is not prognostic (HR 1.42, $P = 0.28$), so there was no evidence that the joint classification improved prognostic fit over the additive one. Adding D1 does improve it: it gives a partial likelihood-ratio statistic of 10.44 on 1 degree of freedom ($P = 0.0012$), an adjusted hazard ratio per standard deviation of 0.82 (0.73–0.92), and a concordance increase from 0.592 to 0.621. Re-coding the two classifiers as a single saturated four-level categorical variable is the same model reparameterized and returns the identical statistic ($\chi^2 = 10.44$). D1's prognostic information is therefore not captured by the saturated four-level joint classification.

## 19 Adjustment for clinical covariates and tumor purity

The association of the frozen DeSurv linear predictor with survival was estimated in cohort-specific Cox models before and after adjustment for the clinical covariates that cohort annotates and, where available, for tumor purity (Table S7). Purity annotation is present in only a subset of cohorts and is derived by different estimators across them (TCGA-PAAD purity was estimated by ABSOLUTE; the PACA-AU and Moffitt purity fields were used as distributed with the cohort annotations, which do not identify the estimator), so it is used here solely as an adjustment covariate; wherever purity could be measured on a full cohort the adjusted hazard ratio was essentially unchanged from the unadjusted one, providing no evidence that the association was explained by measured tumor purity. Consistently, the D1 score itself correlated only weakly with purity in the two full-cohort datasets (Spearman $\rho = 0.06$ in PACA-AU array and $-0.05$ in PACA-AU RNA-seq), the values reported in the main text.

## 20 Empirical optimization stability across random initializations

The idealized alternating scheme is not accompanied by a formal convergence guarantee for the fitted models. With $\lambda_H = 0$, as in the reported analyses, strict convexity of the $H$-subproblem and uniqueness of its block minimizer are not guaranteed a priori, so convergence results that require unique block minimizers do not apply directly; in addition, the implemented algorithm combines column normalization, gradient balancing between the reconstruction and Cox terms, and a Coxnet approximation of the $\beta$ update. We therefore assessed optimization behavior empirically across $R = 100$ independent random initializations of the selected PDAC model, with a stopping tolerance of $10^{-7}$ and a maximum of 4,000 iterations.

100 of the 100 runs terminated on the stopping criterion rather than the iteration cap, none failed numerically, and in every run the recorded penalized objective decreased monotonically to a numerically stable value; the median number of iterations was 38 (range 17–233). The size of the decrease from a random start varies

widely between initializations, but that quantity primarily reflects the starting objective value rather than agreement among the final solutions. Whether independent runs of the reported procedure recover the same gene programs is addressed in Stability of the recovered gene programs below.

## 21 Stability of the recovered gene programs

The previous section documents that the optimizer converges. It does not establish that the programs it converges to are the same ones each time, and that is the property the manuscript relies on when it applies a frozen decomposition across cohorts. We therefore measured stability at the level of the recovered programs rather than the objective, using the same $R = 100$ cached initializations. No model was refitted for the restart analysis.

Individual restarts are highly dispersed. Comparing pairs of independent random restarts, and matching their columns one to one by absolute Spearman correlation of gene loadings, the median overlap between the top 270 genes of matched programs was 0.13 by Jaccard index, against a chance baseline of 0.074 for two independent draws of that size from 1,970 genes. Matched gene-loading correlations were moderate (0.59–0.60 across the three programs). Over the same runs, training concordance was nearly constant (median 0.704, standard deviation 0.010). Near-identical predictive performance from bases that share little gene membership is consistent with a non-identifiable objective, and it is precisely why the reported model is not a single run.

The reported procedure is reproducible with respect to initialization. The reported basis is obtained by aggregating restarts into a consensus initialization and then fitting from it. To test whether that procedure returns the same programs, we partitioned the cached restarts into two disjoint halves, built a consensus initialization from each half independently, fitted from each at the production settings, and compared the results (8 random splits). Across the eight random splits, the paired consensus fits agreed closely with each other (gene-loading correlation 0.93–0.95, patient-score correlation 0.97–0.99), and each agreed still more closely with the reported basis (gene-loading correlation 0.97–0.99, top-270 Jaccard 0.73–0.88). Splitting the restarts halves the number aggregated relative to the reported procedure, so these figures are conservative.

The fitted basis also moved materially from the consensus initialization $W_0$: with programs corresponding by position (confirmed against the full correlation matrix), gene-loading correlations between $W_0$ and the final $W$ were 0.57–0.81 and top-270 Jaccard overlaps were 0.44–0.82. The consensus therefore supplies a reproducible initialization rather than fixing the final basis.

The optimizer alone is not identifiable. The consensus step is what makes the reported decomposition reproducible rather than one draw from a wide distribution, and reproducibility claims in this manuscript refer to that end-to-end procedure.

Stability under data perturbation is a separate and weaker property. The analyses above vary the initialization while holding the data fixed. Refitting on 25 repeated 80% subsamples of the training cohort, with each DeSurv refit a single run from one random initialization at the production hyperparameters rather than the 100-restart consensus procedure, and comparing the top-100 gene sets of the D1-matched factor across subsamples instead gives a mean pairwise Jaccard of 0.34 for DeSurv, against 0.49 for Cox partial least squares, 0.38 for supervised principal components and 0.22 for sparse Cox. DeSurv is mid-range. The 100-gene list is shorter than the 270-gene projection support so that the comparison is on a common footing with the single-vector comparators, whose gene rankings are not factor-specific. This comparison favors dense methods by construction, because a score that spreads weight across many genes has a more stable top-100 list than one built from a selective basis, as the sparse Cox value illustrates; it also disfavors DeSurv relative to the consensus procedure used for the reported model, because each refit here is a single run. Gene-level resampling stability addresses a complementary property distinct from reproducibility of the frozen end-to-end procedure.

## 22 Simulation results under null and mixed scenarios

To characterize the operating characteristics of DeSurv across different signal regimes, we evaluated performance under three simulation scenarios. In the prognostic scenario (main text Fig. 5), the factor-specific survival signal contributed relatively little transcriptomic variation compared with the shared outcome-neutral background signal, and survival depended on 150 factor-specific marker genes ($\beta_1 = 2$, $\beta_2 = \beta_3 = 0$). In the null scenario, survival times were generated independently of gene expression ($\beta = 0$), so no prognostic structure existed. In the mixed scenario, survival depended on 300 genes per survival-driving factor, of which 50% were factor-specific markers and 50% were background genes with loadings shared across all factors. All scenarios used $p = 3{,}000$ genes, $n = 200$ samples, $k = 3$ true factors, and identical gamma distribution parameters for the gene loading matrix $W$ (see Methods). Performance was evaluated using the concordance index (C-index) on the held-out 30% test split of each replicate, precision of recovered prognostic gene sets, and the distribution of selected ranks across simulation replicates.

In the prognostic scenario, median matched-factor precision was 0.50 for DeSurv versus 0.07 for unsupervised NMF, and global marker recall, the fraction of the survival-driving marker genes recovered anywhere in the union of all learned factors' top-$n_{\text{top}}$ gene sets, was 0.26 versus 0.15. Recall is insensitive to factor matching because it searches the union of all learned factors rather than a single matched factor; like precision, it depends on each method's selected $n_{\text{top}}$. It complements the matched-factor precision reported in main-text Fig. 5.

Under the null scenario (Fig. S3a–b), DeSurv showed no advantage over unsupervised NMF ($\alpha = 0$): both methods yielded C-index values near 0.5, and selected ranks concentrated at low values (NMF favoring $k = 2$), consistent with the absence of survival signal. Because the survival objective is uninformative under the null, the cross-validated concordance is essentially flat in $\alpha$ and the Bayesian-optimized supervision strength is unidentified, dispersing across its admissible range rather than concentrating near zero (median $\alpha = 0.59$, IQR 0.19–0.82; range 0–0.94 across 100 replicates). The null-scenario specificity thus reflects the flat concordance surface itself, not the optimizer driving $\alpha$ to zero.

Under the mixed scenario (Fig. S3c–f), DeSurv's advantage was concentrated in marker-gene recovery. Precision against the factor-specific marker genes alone (150 genes; Fig. S3d left) substantially favored DeSurv, whereas precision against the full 300-gene survival set including background genes (Fig. S3d right) was more modestly improved, indicating similar background-gene recovery between methods. The learned factor aligned with the true marker program also received a larger Cox coefficient ($|\beta|$) under DeSurv (Fig. S3e), and C-index was modestly higher (Fig. S3c). Across the three scenarios the pattern was graded: DeSurv's advantage was greatest when variance and prognosis diverged (prognostic), attenuated when they partially overlapped (mixed, as background-gene dilution reduced the discrimination gain), and absent when no survival signal existed (null). These simulations indicate that DeSurv is most advantageous when prognostic structure is not aligned with the dominant sources of transcriptomic variation.

## 23 Cross-validated C-index across factorization rank

To evaluate how the number of latent factors $k$ affects prognostic performance, as a sensitivity analysis we performed an exhaustive grid search over $k \in \{2, 3, \ldots, 12\}$ and supervision strength $\alpha \in \{0, 0.05, \ldots, 0.95\}$ using 5-fold cross-validation on the TCGA+CPTAC training cohort. For each $k$, the best-performing $\alpha$ was selected by maximizing the mean CV C-index. The unsupervised limit ($\alpha = 0$) was evaluated at the same $k$ values for comparison. Results are shown for two gene-selection settings: $n_{\text{top}} = 270$ top genes per factor and $n_{\text{top}} = \text{ALL}$ (all genes retained), both at $\lambda = 0.349$ and $\xi = 0.056$. External validation C-index is shown descriptively; validation outcomes were not used to select $k$, $\alpha$, $\lambda$, $\xi$, or $n_{\text{top}}$.

Fig. S4 shows the resulting C-index curves for both training (CV) and external validation (pooled across held-out cohorts), with shaded ribbons indicating $\pm 1$ standard error. Rank was selected on the training cross-validated concordance. For DeSurv that curve is close to flat across the whole range examined, spanning about two standard errors from $k = 2$ to $k = 12$, with its nominal maximum at $k = 9$ rather than at the selected rank. This fixed-$\lambda$, $\xi$, $n_{\text{top}}$ grid is a different exercise from the full Bayesian-optimization search over all tuned hyperparameters, whose highest observed mean C-index occurred at $k = 7$; applying the same one-standard-error rule to this fixed-hyperparameter sensitivity grid also selects $k = 3$. The selection depends on the gene-selection setting: under panel (b), with all genes retained, the same rule selects $k = 2$. On the training cross-validated curve DeSurv had a higher point estimate than its unsupervised limit at every $k$ examined. The external-validation curve, shown only descriptively, does not show that ordering uniformly: the unsupervised limit is higher at $k = 6$ and at $k = 10$–$12$, and the supervised curve dips at $k = 6$, where the single grid fit at its selected $\alpha$ is the same fit whose factors correspond to almost no reference program (Reference-program coverage, edges and multiplicity). These rank-wise points are single fits, not consensus fits. At the selected rank the validation curve is consistent with the training-based choice, with DeSurv at $k = 3$ attaining concordance comparable to the rank-optimized unsupervised control ($k = 7$; Table S8).

# 24 Standard NMF rank selection heuristics

Standard unsupervised rank selection criteria (reconstruction residuals, cophenetic correlation coefficient, and mean silhouette width) yielded inconsistent guidance for selecting the factorization rank $k$ in the PDAC training cohort (TCGA + CPTAC). Reconstruction residuals decreased smoothly without a visually distinct elbow (Fig. S5a); the Kneedle elbow rule (maximum deviation of the normalized curve from its chord) applied to this curve nonetheless returns $k = 5$, which is the rank used for the elbow-selected Lee NMF comparator in Table S8. Cophenetic correlation fluctuated without a distinct transition point (Fig. S5b). Mean silhouette width favored very low ranks that conflicted with other criteria (Fig. S5c). These inconsistencies illustrate the ambiguity of rank selection under unsupervised criteria in this dataset; the main text selects rank on cross-validated prognostic performance.

# 25 Factor structure of the rank-optimized unsupervised control ($k = 7$)

Throughout the main text, both DeSurv and Lee NMF were fit at the same rank ($k = 3$), enabling a controlled comparison of the effect of survival supervision on factor structure. When the unsupervised limit of the same pipeline selects its own rank, Bayesian optimization at $\alpha = 0$ selects $k = 7$ (the rank-optimized unsupervised control), whereas the one-standard-error rule applied to the supervised DeSurv BO results selects $k = 3$. We therefore examine the factor structure of this independently selected $k = 7$ solution to characterize how the additional factors relate to established programs when supervision is absent (Fig. S6).

At BO-selected $k = 7$ under $\alpha = 0$ (Fig. S6), the rank-optimized unsupervised control corresponds to a broader set of PDAC gene signatures than the matched-rank $k = 3$ solution, with some programs represented by more than one factor; Supplementary Fig. S7 quantifies the accompanying changes in coverage and multiplicity.

## 25.1 Reference-program coverage, edges and multiplicity

Fig. S6 (and main-text Fig. 2a,b) display factor-to-reference-program correspondence as a heatmap. We additionally summarize this same correspondence across factorization rank $k$ using three quantities computed from the same correlation matrix (top 50 genes per factor by factor-specificity score, Spearman correlation of factor loadings against binary reference-program membership, $\rho > 0.2$ representation threshold; Supplementary Information, Section 8), over the fixed 45-program reference universe. Coverage is the number of reference programs with at least one factor at $\rho > 0.2$. Edges is the total number of (factor, reference-program) pairs at $\rho > 0.2$. Reference-program multiplicity is edges divided by coverage, defined only when coverage $> 0$, and reports edges per represented program. A low multiplicity is not by itself evidence that fewer programs are represented, so multiplicity is always reported alongside coverage rather than in place of it.

At each rank $k = 2, \ldots, 12$, these quantities are computed for two single fits at fixed $\lambda$, $\xi$ and $n_{\text{top}} = 270$ from the $k \times \alpha$ cross-validation grid: the unsupervised fit ($\alpha = 0$) and the fit at the $\alpha$ maximizing mean cross-validated C-index, the same $\alpha$-selection convention used for Supplementary Fig. S4. These rank-wise points are single fits at fixed hyperparameters, not the 100-restart consensus fits used elsewhere in the manuscript. Three consensus-fit anchors are overlaid: DeSurv ($k = 3$), standard NMF at matched rank ($k = 3$; Lee NMF), and the rank-optimized unsupervised control ($k = 7$, $\alpha = 0$).

At these anchors, DeSurv reaches coverage 16, edges 16, multiplicity 1.00; matched-rank standard NMF reaches coverage 15, edges 15, multiplicity 1.00; and the rank-optimized unsupervised control at $k = 7$ reaches coverage 32, edges 36, multiplicity 1.125.

The supervised curve collapses at two ranks: at $k = 6$ ($\alpha = 0.1$) and $k = 8$ ($\alpha = 0.2$) the single grid fit reaches coverage 2 and 2 (edges 2 and 3), because the largest correlation between any of its factors and any reference program is only 0.21 and 0.29, barely above the 0.2 representation threshold, whereas the unsupervised fits at the same ranks and the supervised fits at the neighboring ranks $k = 5, 7, 9$ all reach at least 0.57. These are single fits from one random start at fixed hyperparameters, and the $k = 6$ fit is also the one whose external concordance dips in Supplementary Fig. S4; their collapse is the single-run dispersion documented in Stability of the recovered gene programs, not a property of the consensus fits. Across the remaining ranks, coverage increases with rank considerably more than multiplicity does, indicating that the additional factors at higher rank extend correspondence to more reference programs rather than densely duplicating correspondence to the same ones. Through $k = 7$ the dominant change is increased coverage with modest repeated representation. This is not a claim about predictive performance, which is addressed separately (Table 1; Supplementary Table S8).

## 25.2 Stromal resolution: which reference programs are recovered from bulk

Bulk deconvolution has conventionally separated activated from normal stroma, whereas the finer cancer-associated fibroblast states, and the proCAF/restCAF classification derived from them, were established in single-cell data. We therefore asked which stromal reference programs each $k = 3$ factorization actually resolves. Two views are reported because they do not agree, and the defensible claim depends on which question is asked.

In the top-gene view shown in main-text Fig. 2a,b, which asks whether signature genes are represented among the 50 genes that define each factor, the activated-stroma program is recovered comparably by all methods (7, 6 and 10 of 206 genes for DeSurv, standard NMF and the matched-budget control). The restraining-stroma programs behave differently: 2 of 25 restCAF genes and 2 of 35 iCAF genes appear among DeSurv's defining genes, against 0 and 0 for standard NMF. The proCAF programs, by contrast, are recovered comparably by all methods (4 versus 3 of 25 genes).

In the full-gene view, which correlates loadings across all 1,970 analyzed genes with signature membership, no method separates restCAF from proCAF: every correspondence lies between 0.04 and 0.10, whereas the activated-stroma axis is recovered by all methods at 0.19. This view is far more dilute for a 15–25 gene signature among 1,970 genes and correspondingly low powered, but it bounds what can be claimed. We therefore report the restraining-stroma difference as marker-gene presence among the defining genes of each factorization, and do not claim to have resolved the single-cell CAF states from bulk expression.

Note also that the SCISSORS iCAF and myCAF reference signatures are displayed as restCAF and proCAF in main-text Fig. 2, following the naming used by the DeCAF classification; the underlying gene sets are the single-cell-derived ones.

## 25.3 Matched-budget unsupervised control at the same rank

The matched-rank comparison in the main text uses Lee multiplicative NMF, which receives neither DeSurv's consensus initialization nor its hyperparameter tuning. That asymmetry leaves open whether the difference in factor structure reflects tuning effort rather than supervision. We therefore fit an unsupervised factorization control matched to DeSurv on rank, optimizer, search budget, gene filter, cross-validation folds and consensus procedure. With $\alpha$ fixed at 0 and the rank fixed at DeSurv's selected $k = 3$, the remaining hyperparameters were re-optimized within this otherwise identical pipeline, and the resulting model was fit by the same 100-restart consensus procedure used for DeSurv. This comparison isolates removal of survival supervision at the pipeline level rather than holding all non-supervision hyperparameters numerically fixed; the sensitivity analysis below holds DeSurv's $\lambda$, $\xi$ and $n_{\text{top}}$ fixed and varies only $\alpha$.

The control selected $\lambda = 0.0025$, $\xi = 0.0566$ and $n_{\text{top}} = 299$; note that its gene support was selected near the upper bound of the search range (300), whereas the supervised model settled at $n_{\text{top}} = 270$ well inside it. Because $\alpha = 0$ removes the survival term, the selected $\lambda$ is not comparable like-for-like with the supervised model's.

The control does not recover D1. Its best-matching factor reaches Spearman $\rho = 0.24$ against D1, slightly lower than the $\rho = 0.36$ attained by `NMF::nmf` at the same rank, while still recovering D2 ($\rho = 0.86$) and D3 ($\rho = 0.75$). Its external discrimination is likewise no better than `NMF::nmf` at $k = 3$ (Table S8). A sensitivity analysis holding the supervised model's own $\lambda$, $\xi$ and $n_{\text{top}}$ fixed and varying only $\alpha$ gave the same conclusion. Matching the tuning budget therefore does not close the representational gap, supporting survival supervision rather than tuning effort alone as the source of the reorganization reported in the main text.

Lee NMF at elbow-selected $k = 5$ and the rank-optimized unsupervised control at $k = 7$ both substantially improve over Lee NMF at $k = 3$ across all cohorts (Table S8); additional unsupervised factors capture prognostically relevant variation missed at the matched rank. At $k = 7$, the rank-optimized control exceeds DeSurv's C-index in all five validation datasets, so unsupervised factorization given more factors discriminates at least as well as the supervised three-program model. The tradeoff is between that higher discrimination and a less parsimonious factorization: the $k = 7$ control represents a broader set of established programs across seven factors (Fig. S6; Supplementary Fig. S7) rather than concentrating the prognostic signal in three programs. Both the $k = 7$ control and the DeSurv linear predictor retain significant prognostic value after adjustment for established classifiers (main-text Table 1); DeSurv does so with fewer factors.

# 26 Supplementary Data: factor-specific gene lists

As Supplementary Data, Tables S10–S12 list the top 270 genes for each of the three DeSurv factors, ranked by the factor specificity score $s_{ij}$ defined in Supplementary Information, Section 8. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ($n_{\text{top}} = 270$); these are the genes used to compute sample-level factor scores for survival prediction.

Gene symbols in these tables are retained as distributed with the source expression matrices compiled for the training and validation cohorts (Section 6) and were not updated to current HGNC nomenclature; several therefore appear under previous symbols (for example SEPT11, WISP1, CYR61 and H1F0). Because the analysis set was selected for high mean expression as well as high variance, abundantly expressed ribosomal-protein and housekeeping genes are eligible for the factor supports, and some appear among the D2 and D3 defining genes (for example RPS26 and RPS28 in D2; RPS16, RPS21, RPL28 and GAPDH in D3); they enter the supports through the factor-specificity ranking and are not excluded post hoc. In the fitted model, each factor had at least 270 genes with positive factor specificity (284, 401, 1,285 for D1, D2 and D3). Because this specificity can be positive for at most one factor, the three top-270 sets were disjoint, yielding an 810-gene projection support. Factor D1 is enriched for classical tumor and restCAF-like stromal genes, Factor D2 for proCAF-associated stromal genes, and Factor D3 for basal-like tumor genes.

# 27 Translational transportability into independent treated PDAC cohorts

The programs were learned in treatment-naive, non-metastatic PDAC, the setting in which the gene-program matrix $W$ and Cox coefficients $\beta$ were trained. As a translational test we projected the fixed programs, without retraining, into independently treated PDAC and asked whether they retained clinically relevant associations. We used two resources. The first is the O'Kane/COMPASS cohort of treated metastatic PDAC, laser-capture microdissected and therefore epithelial-enriched, which supports a transportability test of the tumor-associated $D_1$ direction under a model stratified by treatment arm (Fig. 4a). The second is the borderline-resectable or locally advanced Linehan cohort, in which paired pre- and post-treatment biopsies allow a within-patient test of whether the projected $D_2$ score changes with therapy (Fig. 4b).

Each analysis is confined to a single cohort; no estimate is pooled across cohorts. Within O'Kane/COMPASS, arm-specific estimates are reported alongside a common $D_1$ slope from a Cox model stratified by treatment arm, which allows arm-specific baseline hazards. The elastic-net penalty shrank the trained survival coefficient for $D_2$ to exactly zero in the treatment-naive model (an estimated selection outcome, not an imposed constraint), so the paired analysis is reported as a change in program level and not as evidence that $D_2$ predicts survival.

### 27.1 Prognostic transportability of D1 in treated metastatic disease

To evaluate transportability in treated metastatic disease while retaining treatment-arm structure, we projected the frozen DeSurv programs into the O'Kane/COMPASS cohort and first estimated $D_1$ associations separately within each arm. Higher $D_1$ scores had protective point estimates in all three treatment arms: FOLFIRINOX (HR per SD 0.80, 95% CI 0.61–1.03), gemcitabine/nab-paclitaxel (HR 0.57, 95% CI 0.37–0.86) and the GA/experimental arm (HR 0.83, 95% CI 0.58–1.20; Fig. 4a). An arm-stratified Cox model allowing treatment-specific baseline hazards yielded a common association of HR 0.75, 95% CI 0.62–0.90 ($P = 0.003$; 173 patients, 123 deaths), with no evidence of heterogeneity across treatment arms (interaction $P = 0.40$). Because these specimens were epithelial-enriched by laser-capture microdissection, this analysis specifically tests transportability of the tumor-associated $D_1$ direction rather than the stromal programs.

### 27.2 Treatment-associated change in D2

In the 29 Linehan patients with paired pre- and post-treatment biopsies, the projected $D_2$ score increased significantly from pre- to post-treatment (paired Wilcoxon $P = 0.013$; mean $\Delta = +0.34$ SD; Fig. 4b), indicating a shift toward proCAF-associated bulk expression. This bulk analysis cannot distinguish a change in fibroblast state from altered tissue composition.

### 27.3 Caveats

These analyses are exploratory and do not establish treatment-predictive effects; treatment, stage and cohort are confounded, the treated cohorts are modestly sized, and no analysis plan was preregistered. The O'Kane/COMPASS analysis is prognostic within treated metastatic disease, whereas the Linehan analysis tests a paired change in program score. Each analysis is confined to a single cohort, and the O'Kane/COMPASS common slope is estimated within that cohort from a model with arm-specific baseline hazards; it is not a pooled estimate across cohorts. We do not test a survival association for $D_2$ in the paired Linehan cohort, because its trained coefficient is zero. No $D_2$ analysis is possible in O'Kane/COMPASS, whose specimens are laser-capture epithelium rather than bulk tissue. Robust treated-cohort validation, and any assessment of whether the programs predict differential treatment benefit, will require larger harmonized series with adequately powered treatment-by-program interaction analyses.

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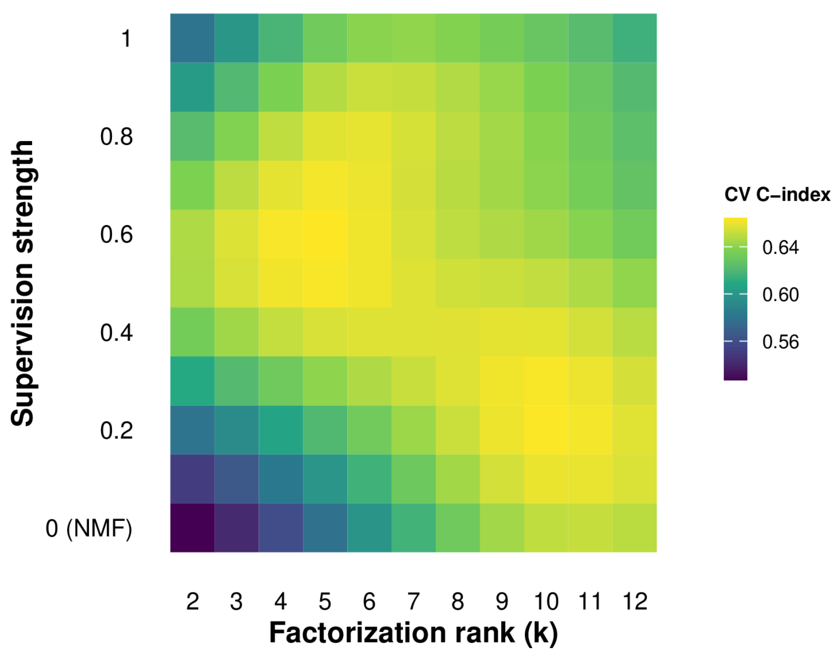


Figure S1: Bayesian-optimization tuning surface for DeSurv in the PDAC training cohort (TCGA + CPTAC). Gaussian-process–predicted mean cross-validated C-index over factorization rank ($k$) and supervision strength ($\alpha$; $\alpha = 0$ corresponds to the unsupervised limit). The response surface shows a broad region of high cross-validated performance at intermediate supervision and moderate rank, a second high region at $k \geq 9$ and $\alpha \approx 0.2$–$0.3$, and discrimination degrading toward the unsupervised ($\alpha = 0$) boundary.

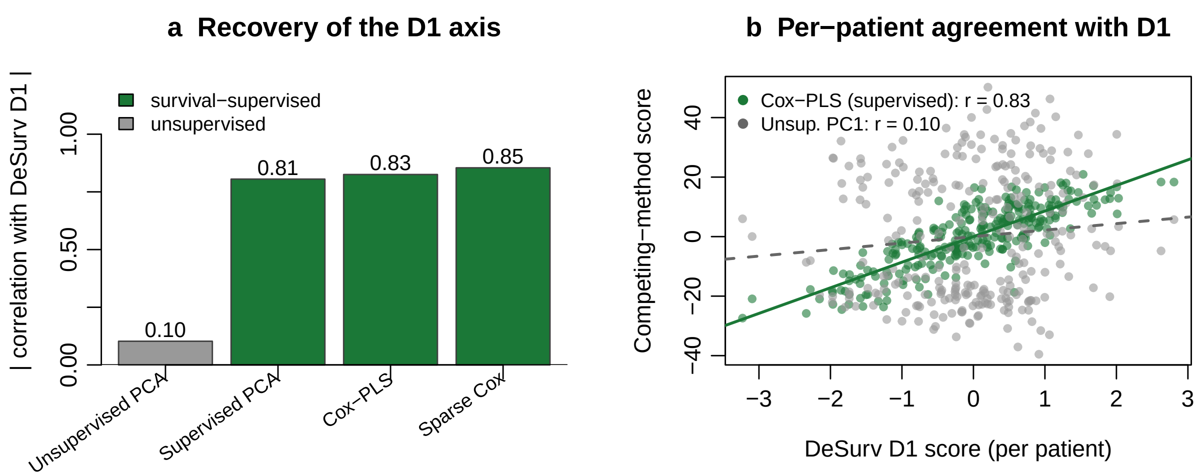


Figure S2: Independent survival-supervised methods recover the D1 axis; an unsupervised method does not. (a) Absolute correlation (across training-cohort patients) of each method's leading direction with the DeSurv D1 score, in the training cohort: supervised principal components, Cox partial least squares, and sparse Cox all recover D1 (green), whereas the leading unsupervised principal component does not (grey). (b) Sample-level agreement with D1 for Cox-PLS (supervised, green) versus the first unsupervised principal component (grey).

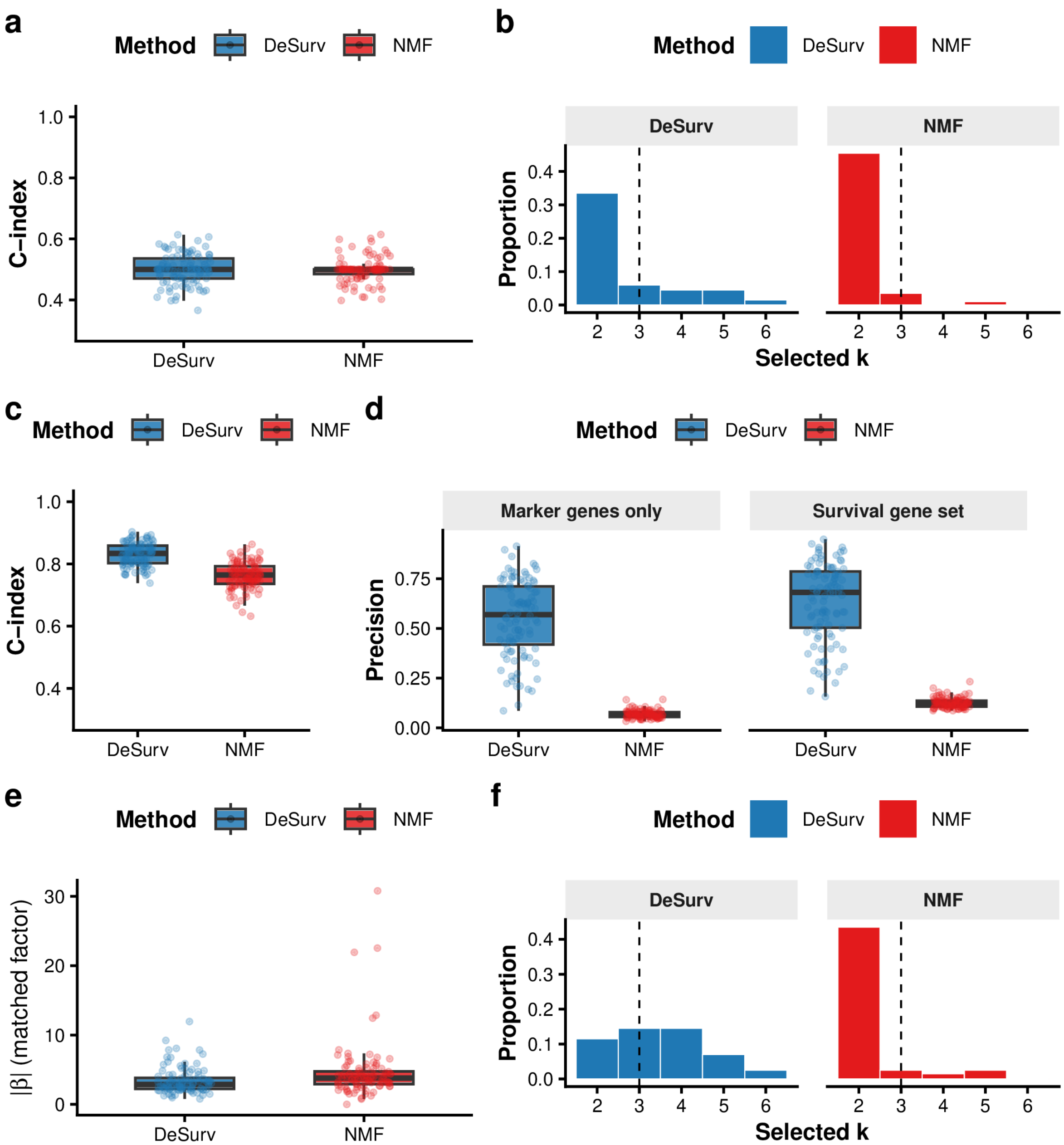


Figure S3: DeSurv performance under null and mixed simulation scenarios. (a,b) Null scenario ($\beta = 0$): both methods yield C-index $\approx 0.5$ (a) and comparable rank distributions (b), so both methods show no out-of-sample prognostic discrimination under the null; precision is omitted as no true survival genes exist. (c–f) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (c) C-index is modestly higher for DeSurv. (d) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (e) The matched factor receives a larger fitted Cox coefficient $|\beta|$ under DeSurv. (f) Selected-rank distributions. DeSurv (blue); NMF (red, $\alpha = 0$); both Bayesian-optimized.

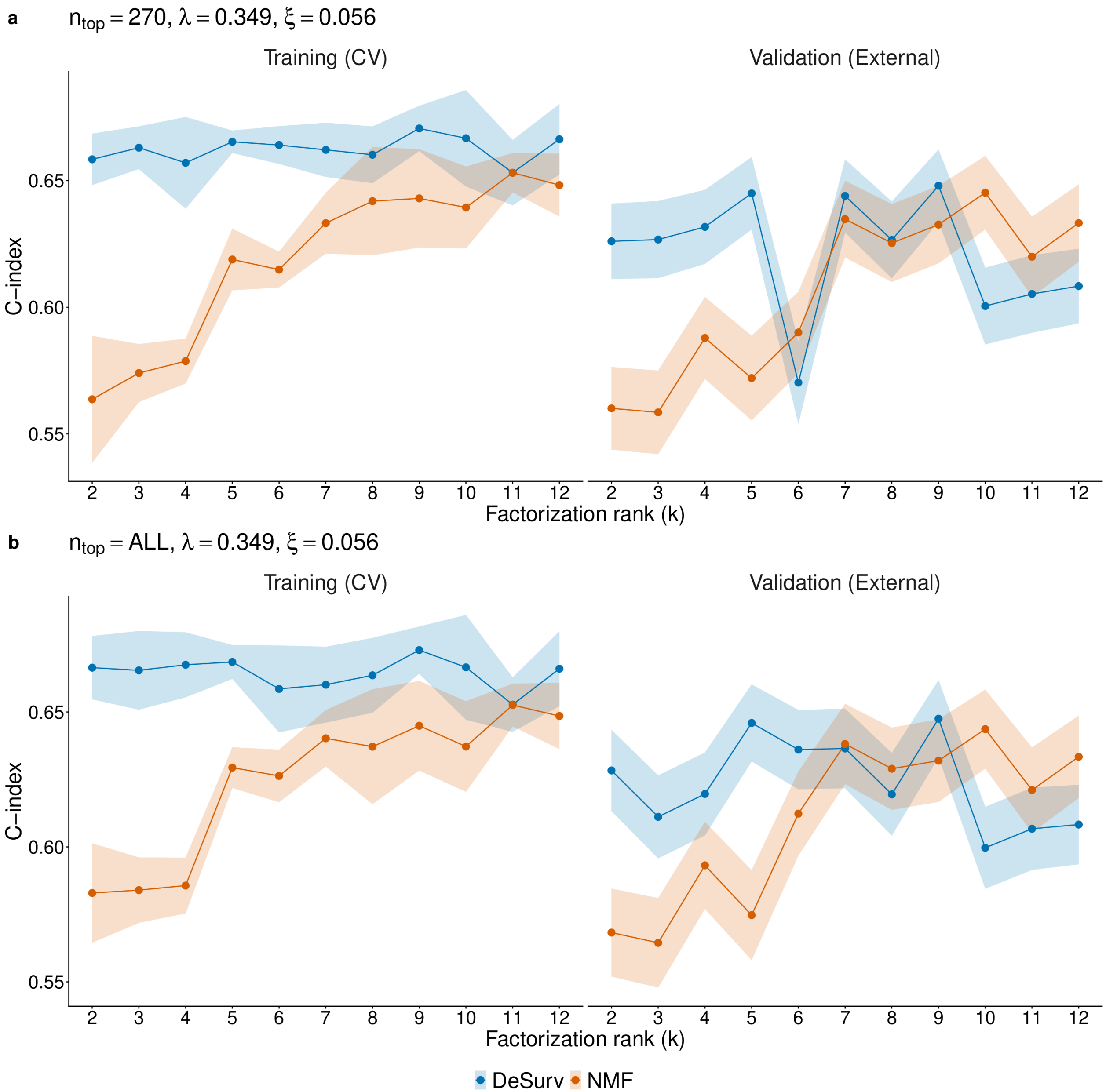


Figure S4: Training (CV) and external validation C-index as a function of factorization rank $k$ for DeSurv (blue) and its unsupervised limit (red, $\alpha = 0$). Shaded ribbons show $\pm 1$ SE. For each $k$, the best $\alpha$ was selected by maximizing the mean CV C-index. Validation C-index is pooled across all external cohorts. (a) Results using $n_{\text{top}} = 270$ top genes per factor ($\lambda = 0.349$, $\xi = 0.056$). (b) Results retaining all genes ($n_{\text{top}} = \text{ALL}$; $\lambda = 0.349$, $\xi = 0.056$).

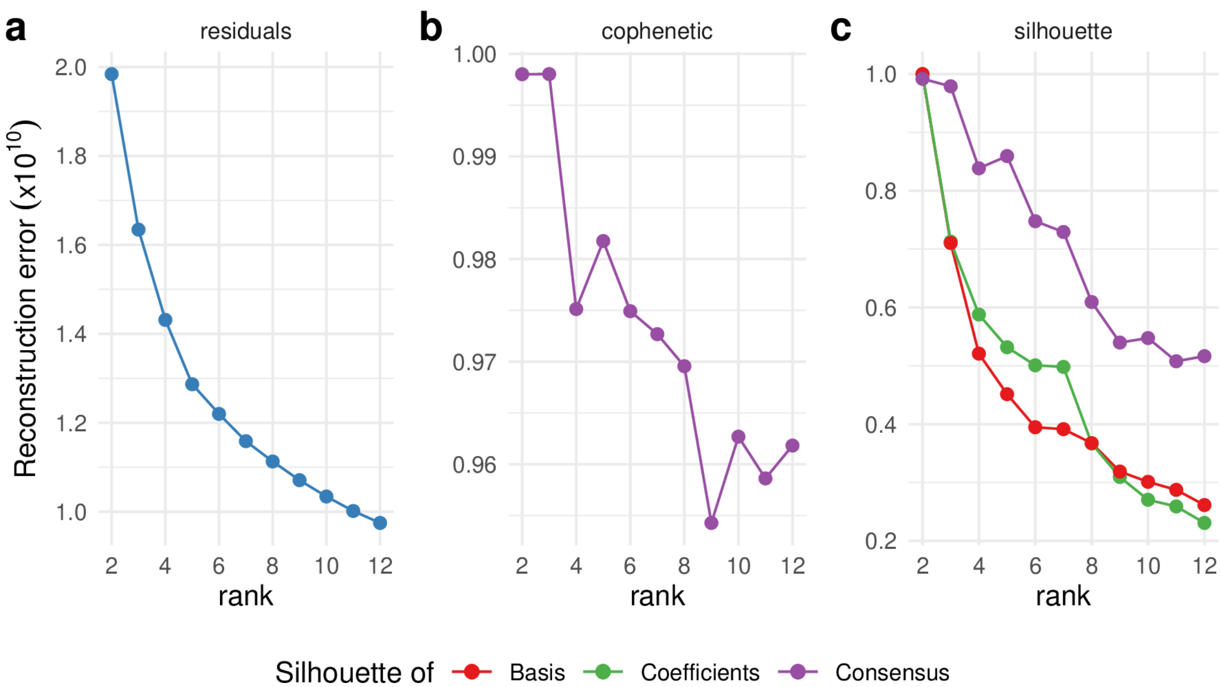


Figure S5: Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (a) Reconstruction residuals as a function of factorization rank $k$. (b) Cophenetic correlation coefficient as a function of $k$. (c) Mean silhouette width of the basis, coefficient and consensus matrices as a function of $k$. These criteria yield inconsistent guidance, illustrating the ambiguity of rank selection under unsupervised criteria; the main text selects rank on cross-validated prognostic performance.

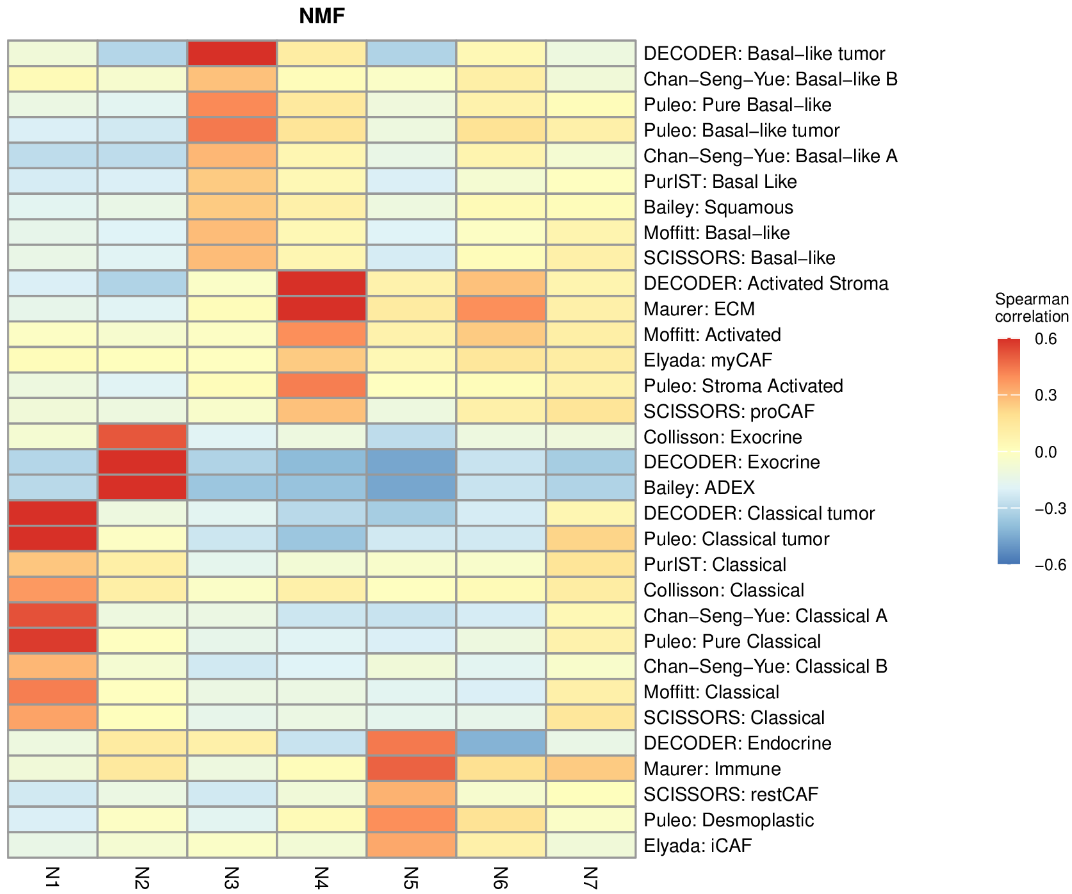


Figure S6: Gene-program correspondence for the rank-optimized unsupervised control ($k = 7$, BO-selected rank at $\alpha = 0$), computed with the same Spearman-correlation analysis and reference programs as main-text Fig. 2a,b (top 50 genes per factor; reference programs with $ho > 0.2$ for at least one factor are shown). Compared with the matched-rank $k = 3$ solution, the $k = 7$ solution shows broader correspondence to established PDAC programs, with some programs represented by more than one factor. Supplementary Fig. S7 quantifies the accompanying changes in coverage and multiplicity.

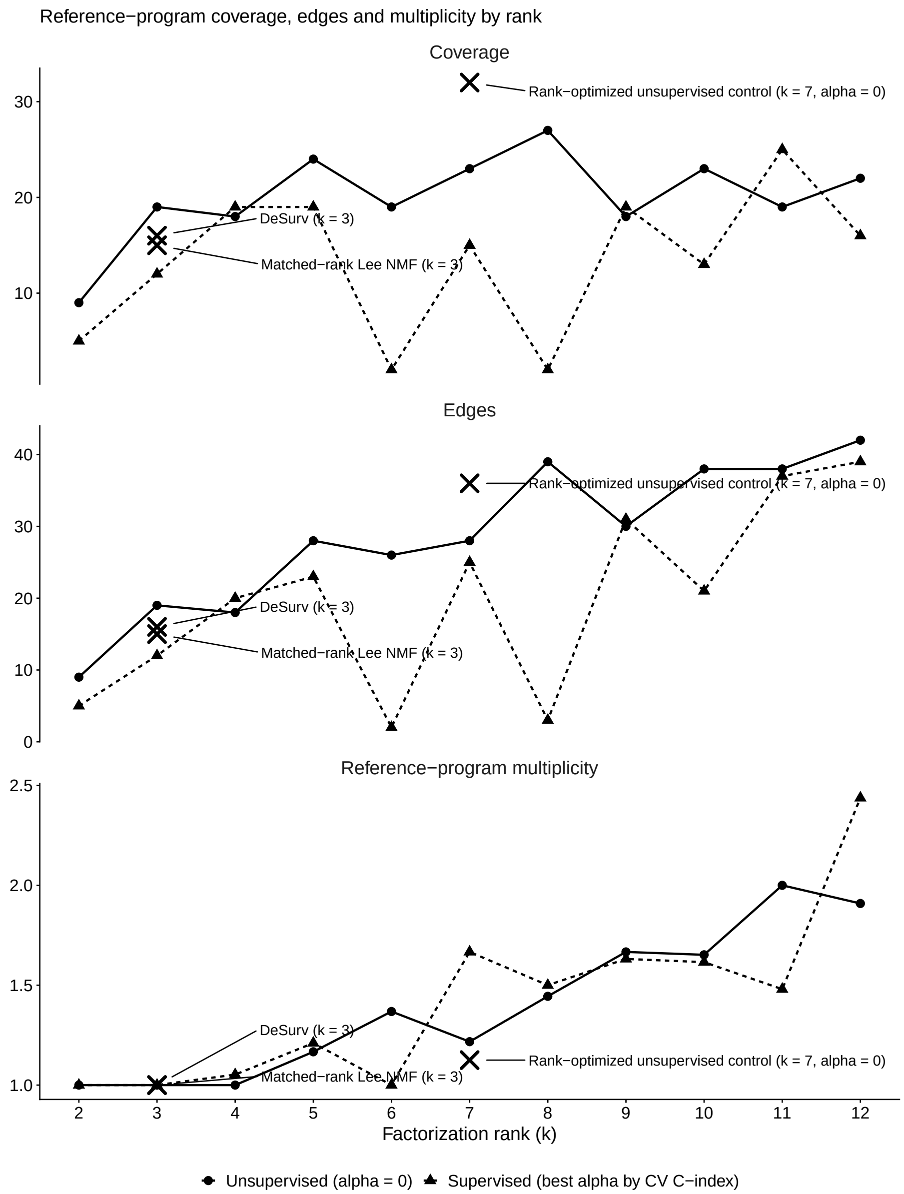


Figure S7: Reference-program coverage, edges and multiplicity by factorization rank $k$, for the unsupervised ($\alpha = 0$) and supervised (best $\alpha$ by cross-validated C-index) curves, computed from the same factor-to-reference-program correspondence analysis as Fig. S6 and main-text Fig. 2a,b (top 50 genes per factor, $ho > 0.2$ threshold, same reference-program universe). Rank-wise points are single fits at fixed $\lambda$, $\xi$ and $n_{\text{top}}$, not the 100-restart consensus fits used elsewhere; markers denote the three consensus-fit anchors (DeSurv $k = 3$, matched-rank standard NMF $k = 3$, rank-optimized unsupervised control $k = 7$).

Table S1: Where survival supervision enters, and what a new sample requires, for DeSurv and five comparators. Cells for DeSurv, standard NMF, supervised principal components and penalized Cox regression are taken from the Methods of this manuscript; CoxNMF cells from its published objective function and SurvNMF cells from its full text; both works are cited in Methods, Hyperparameter selection and cross-validation. "Not described in the source" means the cited work does not state the behavior; no other cell is inferred. The table records architecture and does not rank the methods. Note: CoxNMF does not describe a procedure for scoring new samples. The implication that sample-side loadings would need to be obtained for new samples follows from its published objective and is an inference, not a reported implementation.

| Method | Learned representation | Where survival enters | Gene-program basis directly supervised? | New-sample scoring | External latent-variable re-estimation? |
|---|---|---|---|---|---|
| Standard NMF | Gene programs $W$ and sample loadings $H$ | Not in the factorization; Cox fitted afterwards on projected scores (Methods) | No | In this study, projection onto the fixed $W$, as for DeSurv (Methods) | No (projection) |
| DeSurv (this work) | Gene programs $W$, sample loadings $H$ and Cox coefficients $\beta$ on $Z = W^\top X$ | Cox partial likelihood on $Z = W^\top X$; its gradient acts on $W$ (Methods) | Yes | Projection $Z = \widetilde{W}^\top X_{\text{new}}$ with $\widetilde{W}$ frozen (Methods) | No |
| CoxNMF | Basis $W$ (genes) and coefficients $H$ (samples), with Cox coefficients $\beta$ on $H$ | Cox partial likelihood whose covariates are the columns of $H$ (published objective) | No: the Cox term does not depend on $W$ (published objective) | Not described in the source | Not described in the source |
| SurvNMF | Sample-side latent matrix and feature matrix (the former is denoted $W$ in SurvNMF's transposed notation; full text) | Cox model fitted jointly on the sample-side latent matrix (full text) | No: the Cox term is on the sample side (full text) | Held-out loadings re-solved by constrained least squares (HALS) with the feature matrix fixed (full text) | Yes, for held-out samples; reported for simulated data only (full text) |
| Supervised principal components | Principal components of genes passing a univariate Cox score threshold | Gene screening by Cox score before the components are formed (Methods) | No: components are unsupervised on the screened genes | Fixed linear map over the screened genes (Methods) | No |
| Penalized Cox regression | One coefficient vector over genes (a single linear predictor) | Directly: the objective is the penalized Cox likelihood (Methods) | No gene programs are learned | Fixed linear predictor (Methods) | No |

Table S2: Summary of PDAC cohorts used in this study. $N$ denotes the number of samples after all quality-control filters (histology, survival validity, dataset-specific exclusions). Patient-level metadata: OS = overall survival (time and event indicator); PurIST = tumor subtype classification (Rashid et al. 2020); DeCAF = CAF subtype classification (Peng et al. 2026). Expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study (Peng et al. 2026). Events denotes the number of overall-survival events (deaths) with the corresponding event rate; outcome frequency was lower and more homogeneous in the training cohorts (50–52%) than across the validation cohorts (60–90%, pooled 67%), a case-mix difference relevant to the interpretation of external-validation performance.

| Cohort | Platform | $N$ | Events | Role | Accession | Metadata | Reference |
|---|---|---|---|---|---|---|---|
| TCGA-PAAD | RNA-seq | 144 | 75 (52%) | Training | GDC (TCGA-PAAD) | OS, PurIST, DeCAF | Raphael et al. (2017) |
| CPTAC | RNA-seq | 129 | 64 (50%) | Training | GDC (CPTAC-3) | OS, PurIST, DeCAF | Cao et al. (2021) |
| Dijk | RNA-seq | 90 | 81 (90%) | Validation | E-MTAB-6830 | OS, PurIST, DeCAF | Dijk et al. (2020) |
| Moffitt | Microarray | 123 | 83 (67%) | Validation | GSE71729 | OS, PurIST, DeCAF | Moffitt et al. (2015) |
| PACA-AU (array) | Microarray | 63 | 38 (60%) | Validation | GSE36924 | OS, PurIST, DeCAF | Bailey et al. (2016) |
| PACA-AU (RNA-seq) | RNA-seq | 52 | 31 (60%) | Validation | EGAS00001000154 | OS, PurIST, DeCAF | Bailey et al. (2016) |
| Puleo | Microarray | 288 | 181 (63%) | Validation | E-MTAB-6134 | OS, PurIST, DeCAF | Puleo et al. (2018) |
| Training total | | 273 | 139 (51%) | | | | |
| Validation platform profiles total | | 616 | 414 (67%) | | | | |
| Combined unique-patient validation set | | 570 | 388 (68%) | | | | |

PACA-AU contains 46 patients profiled on both the array and RNA-seq platforms; combined (pooled) analyses represent each patient once, retaining the RNA-seq profile for duplicated patients, consistent with the DeCAF preprocessing rule. Platform-specific per-dataset analyses retain both records.

Table S3: The 45-program reference universe used for every factor-to-program correspondence analysis (main-text Fig. 2a,b; Supplementary Figs. S6 and S7), generated from the same renamed and de-duplicated object the analyses read. Displayed alias is the row label used in the figures; Genes is the size of the published list; In analysis set is the number of those genes among the 1,970 analyzed genes, over which the Spearman correlation is computed. The SCISSORS restCAF and proCAF rows are the SCISSORS iCAF and myCAF signatures under DeCAF nomenclature. The two Puleo basal-like rows are different programs: the whole-tumor pure basal-like component and the tumor-compartment basal-like signature.

| Displayed alias | Source | Genes | In analysis set |
|---|---|---:|---:|
| PurIST: Basal Like | Rashid et al. 2020 | 8 | 7 |
| PurIST: Classical | Rashid et al. 2020 | 8 | 8 |
| DECODER: Basal-like tumor | Peng et al. 2019 | 394 | 139 |
| DECODER: Classical tumor | Peng et al. 2019 | 372 | 166 |
| DECODER: Activated Stroma | Peng et al. 2019 | 206 | 108 |
| DECODER: Normal Stroma | Peng et al. 2019 | 97 | 15 |
| DECODER: Immune | Peng et al. 2019 | 408 | 60 |
| DECODER: Exocrine | Peng et al. 2019 | 113 | 44 |
| DECODER: Endocrine | Peng et al. 2019 | 380 | 61 |
| Moffitt: Basal-like | Moffitt et al. 2015 | 25 | 20 |
| Moffitt: Classical | Moffitt et al. 2015 | 25 | 21 |
| Moffitt: Activated | Moffitt et al. 2015 | 25 | 24 |
| Moffitt: Normal | Moffitt et al. 2015 | 25 | 11 |
| Moffitt: Immune | Moffitt et al. 2015 | 25 | 5 |
| Collisson: Classical | Collisson et al. 2011 | 22 | 22 |
| Collisson: Exocrine | Collisson et al. 2011 | 20 | 19 |
| Collisson: QM | Collisson et al. 2011 | 20 | 12 |
| Bailey: Squamous | Bailey et al. 2016 | 87 | 27 |
| Bailey: Immunogenic | Bailey et al. 2016 | 145 | 22 |
| Bailey: Pancreatic Progenitor | Bailey et al. 2016 | 73 | 7 |
| Bailey: ADEX | Bailey et al. 2016 | 57 | 37 |
| Chan-Seng-Yue: Basal-like A | Chan-Seng-Yue et al. 2020 | 100 | 34 |
| Chan-Seng-Yue: Basal-like B | Chan-Seng-Yue et al. 2020 | 100 | 28 |
| Chan-Seng-Yue: Classical A | Chan-Seng-Yue et al. 2020 | 100 | 69 |
| Chan-Seng-Yue: Classical B | Chan-Seng-Yue et al. 2020 | 100 | 31 |
| Maurer: ECM | Maurer et al. 2019 | 222 | 139 |
| Maurer: Immune | Maurer et al. 2019 | 251 | 101 |
| Puleo: Pure Classical | Puleo et al. 2018 | 90 | 71 |
| Puleo: Immune Classical | Puleo et al. 2018 | 56 | 7 |
| Puleo: Desmoplastic | Puleo et al. 2018 | 101 | 65 |
| Puleo: Stroma Activated | Puleo et al. 2018 | 62 | 50 |
| Puleo: Pure Basal-like | Puleo et al. 2018 | 94 | 57 |
| Puleo: Basal-like tumor | Puleo et al. 2018 | 461 | 118 |
| Puleo: Classical tumor | Puleo et al. 2018 | 539 | 273 |
| Elyada: iCAF | Elyada et al. 2019 | 35 | 25 |
| Elyada: myCAF | Elyada et al. 2019 | 16 | 15 |
| SCISSORS: apCAF | Leary et al. 2023 | 25 | 4 |
| SCISSORS: restCAF | Leary et al. 2023 | 25 | 17 |
| SCISSORS: proCAF | Leary et al. 2023 | 25 | 16 |
| SCISSORS: Perivascular | Leary et al. 2023 | 25 | 5 |
| SCISSORS: panCAF | Leary et al. 2023 | 25 | 22 |
| SCISSORS: Basal-like | Leary et al. 2023 | 25 | 24 |
| SCISSORS: Classical | Leary et al. 2023 | 25 | 22 |
| deCAF: proCAF | Peng et al. 2026 | 9 | 4 |
| deCAF: restCAF | Peng et al. 2026 | 9 | 8 |

Table S4: Index of analysis scripts. Scripts are numbered as in the code repository; helper functions shared by several scripts (de-duplication of PACA-AU patients, validation scoring, elbow selection) are called from the scripts listed. Section numbers refer to this document.

| Script | Analysis | Read in |
| --- | --- | --- |
| code/03 | Bayesian optimization of hyperparameters; the matched-budget refinement with $\alpha$ pinned at 0 | Sections 5, 25.3 |
| code/04 | Final 100-restart consensus fits (DeSurv, Lee NMF at matched and elbow rank, unsupervised controls) | Sections 5.1, 21 |
| code/05 | External validation: frozen projection, linear predictor and per-dataset C-index | Sections 7, 17 |
| code/06 | Fixed-hyperparameter $k \times \alpha$ cross-validation grid and its external-validation curve | Sections 23, 25.1 |
| code/07 | Simulation studies (three scenarios, 100 replicates each) | Section 22 |
| code/11 | Treated-cohort analyses (O'Kane/COMPASS arm-stratified D1; Linehan paired D2) | Section 27 |
| code/14 | Variance partition of the D1 score; additive and interaction adjustment for PurIST and DeCAF | Section 18 |
| code/15 | Single-component supervised comparators and recovery of the D1 axis | Section 13 |
| code/16 | Subsample stability of top-gene sets across methods | Section 21 |
| code/17 | Proper-scoring sensitivity (integrated Brier score, IPA) | Section 15 |
| code/18 | Clinical-covariate and tumor-purity adjustment by cohort | Section 19 |
| code/19 | Independently tuned supervised comparators, axis decomposition and compartment attribution | Section 14 |
| code/20 | Dispersion of individual random restarts | Section 21 |
| code/21 | Reproducibility of the consensus endpoint from disjoint restart halves | Section 21 |
| code/23 | Stromal reference-program resolution (top-gene and full-gene views) | Section 25.2 |
| code/24 | Reference-program coverage, edges and multiplicity by rank; the reference-universe table | Sections 8, 25.1 |
| code/25 | Between-cohort heterogeneity, proportional-hazards diagnostic and bootstrap C-index intervals | Sections 16, 17 |
| code/26 | Reconstruction of the gradient-scaling factor $\zeta$ along the production optimization path | Sections 2.3, 21 |

Table S5: Cross-method projection $R^2$: how reconstructable each factor's gene-loading vector is from the other method's full basis (OLS $R^2$). DeSurv retains the bulk of the unsupervised structure (NMF N1 and N3 carry over at $R^2 \geq 0.88$) and reorganizes it: the survival-aligned D1 program is essentially unreconstructable from the NMF basis ($R^2 = 0.09$), while the exocrine NMF factor N2 is poorly reconstructed from the DeSurv basis ($R^2 = 0.38$).

| Factor | $R^2$ from the other method's basis |
|---|---|
| NMF N1 (tumor) | 0.90 |
| NMF N2 (exocrine) | 0.38 |
| NMF N3 (microenviron.) | 0.88 |
| DeSurv D1 (classical/restCAF-like) | 0.09 |
| DeSurv D2 (proCAF-like) | 0.81 |
| DeSurv D3 (basal-like) | 0.75 |

Table S6: Whole-matrix reconstruction $R^2$ across the 1,970-gene analysis matrix for unsupervised NMF versus DeSurv at $k = 3$. Two denominators are shown. The centered $R^2$ is the conventional definition, taken about the grand mean, and is the one to read as fraction of variation reconstructed. The uncentered $R^2$ divides by $\sum X^2$; because the analysis matrix holds within-sample ranks with grand mean $p/2$, the mean alone accounts for about 75% of that denominator, so uncentered values are high for any nonnegative low-rank fit and are reported for completeness. On the centered definition DeSurv gives up 6.0 percentage points of reconstruction relative to NMF (1.5 on the uncentered scale).

| Model | Centered $R^2$ | Uncentered $R^2$ | Residual fraction (centered) |
|---|---|---|---|
| NMF ($k = 3$) | 0.812 | 0.953 | 0.188 |
| DeSurv ($k = 3$) | 0.752 | 0.938 | 0.248 |

Table S7: Hazard ratios (per SD of the DeSurv linear predictor) for overall survival, before and after adjustment for standard clinical covariates and, where annotated, tumor purity, fitted separately in each cohort by complete-case analysis (per-cell $n$ and events shown). The Covariates column lists the clinical variables entered (stage, grade, age, sex, nodal status, or resection margin, as available). After clinical adjustment the DeSurv score remains independently prognostic in Puleo, PACA-AU array and both training cohorts, and is attenuated to non-significance in PACA-AU RNA-seq, Dijk and Moffitt, the three smallest complete-case sets. Where tumor-purity annotation was available, purity adjustment left the estimate essentially unchanged wherever it could be measured on the full cohort (PACA-AU array 1.44 versus 1.45, PACA-AU RNA-seq 1.66 versus 1.65, TCGA 3.50 versus 3.60; unadjusted versus purity-adjusted), providing no evidence that the association was explained by measured tumor purity. The smaller Dijk and Moffitt cohorts are individually underpowered, and their estimates are correspondingly unstable: Moffitt is non-significant both unadjusted and after adjustment, and its purity-adjusted model rests on only $n = 29$ of its 123 patients. Training-cohort (TCGA, CPTAC) hazard ratios are in-sample.

| Cohort | Covariates | Unadjusted | Clinical-adjusted | Purity-adjusted |
|---|---|---|---|---|
| PACA-AU array | stage, grade, age, sex | 1.44 (1.07–1.93), 0.016 ($n$=63, 38 ev) | 1.58 (1.06–2.35), 0.025 ($n$=62, 37 ev) | 1.45 (1.08–1.96), 0.014 ($n$=63, 38 ev) |
| PACA-AU RNA-seq | stage, grade, age, sex | 1.66 (1.15–2.38), 0.006 ($n$=52, 31 ev) | 1.56 (0.95–2.55), 0.077 ($n$=51, 30 ev) | 1.65 (1.15–2.36), 0.006 ($n$=52, 31 ev) |
| Puleo | sex, margin | 1.37 (1.18–1.59), < 0.001 ($n$=288, 181 ev) | 1.33 (1.14–1.55), < 0.001 ($n$=284, 177 ev) | — |
| Dijk | grade, age, sex, nodal | 1.32 (1.06–1.66), 0.015 ($n$=90, 81 ev) | 1.18 (0.92–1.50), 0.189 ($n$=85, 77 ev) | — |
| Moffitt | sex, margin | 1.20 (0.98–1.48), 0.076 ($n$=123, 83 ev) | 1.20 (0.94–1.53), 0.154 ($n$=98, 67 ev) | 0.87 (0.50–1.54), 0.639 ($n$=29, 18 ev) |
| TCGA (train) | stage, grade, age, sex, nodal | 3.50 (2.63–4.66), < 0.001 ($n$=144, 75 ev) | 3.32 (2.40–4.59), < 0.001 ($n$=142, 74 ev) | 3.60 (2.68–4.84), < 0.001 ($n$=143, 74 ev) |
| CPTAC (train) | stage, age, sex, nodal | 3.72 (2.73–5.08), < 0.001 ($n$=129, 64 ev) | 3.59 (2.59–4.97), < 0.001 ($n$=123, 61 ev) | — |

Table S8: External validation C-index by method and validation dataset. DeSurv at the one-standard-error-selected rank ($k = 3$) is compared with Lee NMF at matched rank ($k = 3$) and at elbow-selected rank ($k = 5$), with the matched-budget unsupervised control ($k = 3$, $\alpha = 0$) and with the rank-optimized unsupervised control ($k = 7$, $\alpha = 0$). The unsupervised columns are not all produced by one implementation: the two Lee NMF columns are `NMF::nmf`, whereas the two controls are the DeSurv optimizer run at $\alpha = 0$ with the same consensus-initialization procedure as the supervised model. The matched-budget column is the tuning-budget control: it holds rank, optimizer, consensus-initialization procedure and Bayesian-optimization budget fixed while re-optimizing the remaining hyperparameters with $\alpha = 0$. It does not improve on `NMF::nmf` at the same rank. Higher values indicate better discrimination. For this table only, each method was scored on the union of its own factors' top-$n_{\text{top}}$ gene supports ($n_{\text{top}} = 270$) rather than on per-factor supports.

| Dataset | DeSurv ($k = 3$) | Lee NMF ($k = 3$) | Matched-budget control ($k = 3$, $\alpha = 0$) | Lee NMF ($k = 5$, elbow) | Rank-optimized control ($k = 7$, $\alpha = 0$) |
|---|---|---|---|---|---|
| Dijk | 0.607 | 0.542 | 0.516 | 0.633 | 0.631 |
| Moffitt | 0.562 | 0.521 | 0.516 | 0.552 | 0.578 |
| PACA-AU array | 0.649 | 0.470 | 0.462 | 0.657 | 0.669 |
| PACA-AU RNA-seq | 0.634 | 0.430 | 0.412 | 0.679 | 0.702 |
| Puleo | 0.636 | 0.534 | 0.533 | 0.642 | 0.651 |

Table S9: Cohort-specific external-validation estimates for the frozen DeSurv linear predictor, on the de-duplicated patient-level table (570 unique patients, 388 deaths; PACA-AU represented once, RNA-seq retained). Hazard ratios are per standard deviation of the linear predictor computed within that cohort, the convention used for the interaction test, so they are not comparable to the pooled-standard-deviation hazard ratio in main-text Table 1. C-index confidence intervals are percentile intervals from 1000 stratified bootstrap replicates (seed 1) resampling within cohort; the combined C-index is the event-weighted mean of the four cohort values.

| Cohort | $n$ | Deaths | HR per within-cohort SD (95% CI) | C-index (95% CI) |
|---|---|---|---|---|
| Dijk | 90 | 81 | 1.32 (1.06–1.66) | 0.607 (0.531–0.679) |
| Moffitt | 123 | 83 | 1.20 (0.98–1.48) | 0.562 (0.478–0.639) |
| PACA-AU | 69 | 43 | 1.46 (1.09–1.96) | 0.630 (0.541–0.745) |
| Puleo | 288 | 181 | 1.37 (1.18–1.59) | 0.636 (0.594–0.678) |
| Combined | 570 | 388 | — | 0.613 (0.579–0.646) |

Table S10: Top 270 genes for DeSurv factor D1, classical tumor and restCAF-like stroma, ranked by factor specificity score $s_{ij}$. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ($n_{\text{top}}$).

| **Factor D1 (classical tumor + restCAF-like stroma)** | | | | | |
|---|---|---|---|---|---|
| 1–45 | 46–90 | 91–135 | 136–180 | 181–225 | 226–270 |
| DMBT1 | FGFR2 | GOLGA8A | PTPRN2 | RPL9 | PPP1R12B |
| IGFBP2 | COL9A2 | IL2RG | FCGBP | CD79A | PIP5K1B |
| S100A4 | COL22A1 | KIAA1211 | PLA2G2A | PROM1 | AHCYL2 |
| KRT23 | IRF7 | SFRP1 | FABP4 | SCD | SST |
| LIF | SBNO2 | ADH1B | EPPK1 | TMC5 | CYP3A5 |
| IFI6 | MGAT3 | MFSD2A | CPXM2 | CCL14 | SMAP2 |
| PDLIM3 | SYNM | CLU | KIF1A | PLXNB1 | ZG16B |
| CLDN18 | APOD | CLMN | NGFR | TMEM119 | SERPINF1 |
| MYO15B | SEMA6A | CHPF | NFKBIA | SORBS2 | CLDN3 |
| SYT8 | TNXB | MTMR11 | SPOCK2 | C7 | NR1I2 |
| TFF2 | PTGIS | TTR | ATP2C2 | SLC12A7 | JUND |
| TMPRSS2 | CCL18 | WFDC2 | NBEAL2 | QSOX1 | NCF2 |
| CHGB | ZC3H12A | C2orf72 | CAPN9 | UNC5CL | CELSR1 |
| CST1 | CELSR3 | CCND2 | PTGDS | LONRF2 | MUC17 |
| B3GNT7 | EMB | ARHGAP4 | LSP1 | VSIG2 | INPP5D |
| F5 | KCNJ15 | PKDCC | LAMA5 | ADAMTSL2 | TM4SF5 |
| TNRC18 | SORBS1 | SRPX | FASN | ARHGEF16 | HNF4A |
| IGF2 | PAQR8 | CCL21 | GFRA1 | PCOLCE | CFD |
| ARSD | SLC4A4 | CARD11 | PTPRH | CFC1B | HCLS1 |
| ZDHHC7 | ACSS1 | CTRB2 | FMOD | SLC9A3 | SH3PXD2B |
| LENG8 | ST6GALNAC1 | GOLGA8B | FHL1 | ACACB | PCSK2 |
| ATP9A | DES | IYD | PCDH20 | BCAS1 | HMGCS2 |
| TMPRSS3 | PRUNE2 | CORO1A | CCR7 | RASD1 | FILIP1L |
| CES1 | PLCB4 | SAA2 | CHIT1 | LGALS4 | ETNK1 |
| HOPX | CADPS | CDCA7 | PARM1 | MS4A1 | TNNI2 |
| CNN1 | FMO5 | BOC | ITM2A | CAPN6 | TOX3 |
| MIA | RAP1GAP2 | LTB | NAPSB | SCCPDH | FRZB |
| CRIP2 | GALNT12 | COMP | AGT | PTPRCAP | HABP2 |
| NEURL1B | CCL2 | MANSC1 | FUT3 | VNN1 | GDA |
| CHGA | REG3A | SLC19A3 | SIPA1L2 | PI16 | NCOR2 |
| ACTG2 | ZCCHC24 | STEAP2 | SPINK5 | SEMA7A | DNAH2 |
| CXCL14 | LTF | FOXA1 | ACSL1 | IL10RA | DOK4 |
| TPPP3 | EFHD2 | WNK2 | BTNL8 | GALNT3 | HOXB9 |
| PIGR | ATP11A | SELL | CORO2A | GOLM1 | KIAA1324 |
| CYP27A1 | GP2 | PLEKHA6 | LAD1 | TNFAIP2 | TRIM2 |
| ADAM8 | HIPK2 | NFASC | TMEM63A | CD79B | FGFBP1 |
| FGFR3 | SLC26A9 | EGR3 | GC | ENPP2 | IRX3 |
| IGF1R | TACC2 | PLIN4 | C1orf21 | SQLE | TINAGL1 |
| PLEKHB1 | TRAK1 | WDR72 | MIF | KLF4 | SFRP4 |
| CPXM1 | SAA1 | AMY2A | FOXA2 | PDLIM7 | APLNR |
| ATP7B | CLDN23 | IL16 | CCL19 | FOSB | AGRN |
| SLC12A2 | FOXJ1 | TJP3 | GSTA1 | RHPN2 | SLC25A25 |
| CCL20 | FRY | SLC9A2 | LLGL2 | SCG2 | PODN |
| AKNA | GATA6 | LMOD1 | ACP5 | EMILIN1 | ITIH2 |
| COL16A1 | PLXNA2 | CPS1 | UGT2B15 | BACE2 | FLJ23867 |

Table S11: Top 270 genes for DeSurv factor D2, proCAF-like stroma, ranked by factor specificity score $s_{ij}$.

| **Factor D2 (proCAF-like stroma)** | | | | | |
|---|---|---|---|---|---|
| 1–45 | 46–90 | 91–135 | 136–180 | 181–225 | 226–270 |
| UHMK1 | RPS26 | SAMD9L | ATP8A1 | IRF6 | RASSF2 |
| DDR2 | MYLK | DOCK2 | TGFBR1 | KLHL6 | ZNF83 |
| INHBA | COL11A1 | IGF1 | SYNPO2 | C1QTNF3 | CYTIP |
| DYNC2H1 | NFIB | TACC1 | PDE4B | PCLO | SRPX2 |
| ZDHHC20 | PTPN13 | CFH | SGCD | ITGA2 | LY75 |
| ITPR2 | PDE3A | DMD | STK17B | LRIG3 | HLA-DOA |
| HMCN1 | TRIM22 | PCDH7 | NUAK1 | SSPN | STK38L |
| LAMA2 | DPYD | MALAT1 | FCGR3A | F2R | RSAD2 |
| SLFN5 | COL14A1 | NOTCH2 | GPNMB | PDGFC | HBB |
| PLXNC1 | CNTN1 | CDK6 | VCAM1 | CD109 | HERC2P2 |
| SVEP1 | NCKAP1L | FBN1 | ADAMTS9 | CLIC4 | AOX1 |
| CYBB | PHLDB2 | SYNE2 | ITGB8 | SORL1 | PLIN2 |
| SYNE1 | ITGA4 | RHOBTB3 | CR1 | AOAH | PFKFB2 |
| SAMHD1 | MACC1 | CALCRL | SEPT11 | FMO2 | OGN |
| SKIL | SEMA5A | NRP1 | IFI44L | RDX | DOCK10 |
| ADAMTS12 | MRC1 | LIFR | ELL2 | IL1RAP | RGS4 |
| BICC1 | SLIT2 | STAB1 | GPR183 | GBP1 | AXL |
| ITGA1 | ZEB2 | FLRT2 | ZBTB16 | CXCR4 | FGD6 |
| COL8A1 | ADAM10 | RNF144A | FRMD6 | WNK1 | CDC42BPA |
| PDGFRA | SAMD9 | BCAT1 | PDE5A | MFAP5 | SLC6A6 |
| ABCA1 | MS4A7 | CILP | CSF1R | ALDH1A3 | RGS1 |
| TCF4 | EFEMP1 | CCDC80 | PLEK | CD53 | KIAA1324L |
| CEP170 | BIRC3 | FAT4 | SLK | OMD | CPA3 |
| CD163 | WISP1 | BNC2 | SLC7A2 | PLOD2 | CHL1 |
| MAN1A1 | LAMA4 | LTBP1 | HEG1 | ITGAV | SLC7A11 |
| MSR1 | FAM198B | COL12A1 | GREM1 | MACF1 | GFPT2 |
| CCDC88A | PTPRC | IL1R1 | VGLL3 | NID2 | RAI14 |
| F13A1 | FAP | SLA | GBP5 | NAMPT | LMAN1 |
| RPS28 | ARRDC3 | MBOAT2 | SLC1A3 | ADAM28 | MECOM |
| RALGAPA2 | MYH10 | FNDC3B | FGF7 | GNE | FPR1 |
| ZEB1 | FCGR2A | PRRX1 | ABCA8 | ENAH | EDEM3 |
| FBXO32 | PARP14 | COL10A1 | ADAM12 | FAM129A | RASSF6 |
| INPP4B | DSE | DDX60 | CYBRD1 | ECM2 | IQGAP2 |
| CDH11 | EHF | DOCK9 | TFRC | VSIG4 | MIAT |
| PKHD1 | OSMR | SRGN | CTSK | MTAP | PRICKLE1 |
| AKAP2 | ARID5B | FOXO1 | COLEC12 | INMT | CHST15 |
| CYP1B1 | ASPN | MS4A6A | ALDH1L2 | SGMS2 | GEM |
| CCL5 | SLCO2B1 | ADAMTS2 | PGM2L1 | COL4A4 | CHRDL1 |
| FGL2 | FPR3 | CPVL | FAM135A | CAPRIN2 | DSC2 |
| MAP1B | DOCK8 | CD93 | DOCK11 | CENPF | RAB31 |
| PLXDC2 | SLFN11 | DST | EPHA3 | DCLK1 | SLC38A1 |
| LRRK2 | ZNF532 | CDK14 | PAPPA | EDNRA | CNN3 |
| KIAA0754 | SPON1 | MSRB3 | EGFR | WNT5A | FLT1 |
| EDIL3 | SEMA3C | RNF213 | RASA1 | GALNT5 | TMEM200A |
| ITGBL1 | FERMT2 | SEL1L | AHR | SLC38A2 | SLC2A3 |

Table S12: Top 270 genes for DeSurv factor D3, basal-like tumor, ranked by factor specificity score $s_{ij}$.

| **Factor D3 (basal-like tumor)** | | | | | |
|---|---|---|---|---|---|
| 1–45 | 46–90 | 91–135 | 136–180 | 181–225 | 226–270 |
| HSPB1 | HLA-H | LAMB3 | MYL9 | IL1RN | SDC1 |
| KRT7 | HSPA1A | FSCN1 | COL17A1 | TST | KLK11 |
| MSLN | LGALS3 | TSPAN13 | EPS8L1 | RAPGEFL1 | PERP |
| RPSAP58 | FAM83H | TSPAN8 | CDH3 | C15orf48 | ARHGAP23 |
| S100P | EPB41L1 | ALDH3B1 | RNASET2 | PFKP | HPGD |
| RPS16 | EFNA1 | ANPEP | TUBA1C | STARD10 | AKR1C3 |
| IL32 | TKT | GSTP1 | RAP1GAP | FOXQ1 | CRABP2 |
| SH3BGRL3 | SLC2A1 | KLK10 | H19 | REG4 | GRB7 |
| IFITM2 | PLOD1 | B4GALT1 | LTBP3 | GJB1 | SERPINA4 |
| RPS21 | IER3 | GAS6 | AQP3 | SLC29A1 | SLC39A4 |
| SEMA4B | B3GNT3 | ABCC3 | MPZL2 | NOTCH3 | FA2H |
| APOL1 | PRSS3 | CLDN2 | ID1 | TRIM16 | RPL8 |
| S100A16 | TOB1 | CEACAM6 | RARRES2 | TAP2 | MT1X |
| DNAJB1 | EGR1 | CREB3L1 | SLC25A6 | KRT19 | TCN1 |
| MT2A | PLA2G16 | MST1R | RBP1 | CRP | AKAP1 |
| TNFRSF21 | OCIAD2 | CEBPB | HSPA5 | SNHG5 | GJB3 |
| RPL28 | LPCAT4 | TXN | PER1 | KRT18 | DUOXA2 |
| FXYD3 | BST2 | NR4A1 | ADAP1 | CA12 | HLA-DRB5 |
| SERPING1 | EPHA2 | AP1M2 | TMC4 | MYEOV | ARRDC2 |
| HMGA1 | MUC5B | SYNPO | HSD17B2 | P4HB | OAS1 |
| S100A14 | KCNN4 | ANO1 | PTP4A1 | VILL | ADRA2A |
| H1F0 | ITGA3 | SIK1 | PLAUR | HKDC1 | NQO1 |
| KRT17 | TSTA3 | EFNB2 | ANXA10 | STEAP3 | LRG1 |
| CSTB | IFI27 | CDKN1A | PYGB | PLEK2 | SERPINH1 |
| ASS1 | MMP1 | S100A10 | CYP2S1 | CD68 | COL18A1 |
| EFNB1 | SLPI | PKP3 | ALDOA | MAFF | CERCAM |
| MDK | CLTB | MCM7 | CDCP1 | CD9 | CA9 |
| LY6E | SPINK1 | RARRES3 | IFITM3 | PLLP | CDHR2 |
| CAMK2N1 | JUP | CEACAM1 | BAIAP2L1 | EPHB4 | ESRP1 |
| MGLL | PPP1R15A | ALDH2 | MGST1 | KRT16 | CRYZ |
| PDZK1IP1 | TRIM29 | CES2 | TFF1 | ABLIM3 | TAGLN2 |
| BCL2L1 | PPP1R1B | SLC6A8 | MUC4 | PTPRU | HNF1B |
| C19orf33 | ARPC1A | PPL | MMP28 | MUC6 | BMP4 |
| SFN | GPRC5A | NDRG1 | CLDN1 | FDFT1 | MT1E |
| ITGB4 | RAB25 | AZGP1 | TM4SF1 | GAPDH | ANGPTL4 |
| TSPAN1 | ZFP36L2 | S100A11 | TFF3 | EPS8L3 | CFB |
| SLC16A3 | CLDN7 | SEMA3B | UNC13D | CLDN10 | GDF15 |
| C4A | DDIT4 | TNFRSF12A | SERINC2 | ITPKC | PADI1 |
| KLF5 | FKBP4 | SOX9 | SPNS2 | LOXL2 | TMSB10 |
| IER2 | LAPTM4B | PMEPA1 | AQP5 | AMIGO2 | B2M |
| ZFP36L1 | MUC13 | CYR61 | IGFBP4 | DUOX2 | CAV2 |
| MAL2 | PSCA | EPCAM | SLC7A5 | CLDN4 | CTSH |
| SCNN1A | SDCBP2 | BHLHE40 | TESC | CAPG | PABPC1 |
| MALL | PROM2 | CX3CL1 | IFITM1 | CEACAM5 | KRT8 |
| BGN | LRRC8A | SMAD3 | KCNK5 | VWA1 | TAP1 |